

Unit 4

Module 4 : Complex Variables (Differentiation)

5

Complex Variables (Differentiation)

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Syllabus

Differentiation, Cauchy-Riemann equations, analytic functions, harmonic functions, finding harmonic conjugate; analyticity of elementary functions (exponential, trigonometric, logarithm) and their properties;

5.1 Introduction

Complex numbers can be defined as ordered pairs (x, y) of real numbers.

In general, a number of the form $z = x + iy$ is called rectangular form of a complex number where x and y are real numbers while $i^2 = -1$.

x is called the real part of z and is denoted by $\text{Re}(z)$. y is called the imaginary part of z and is denoted by $\text{Im}(z)$.

$$x = \text{Re}(z), y = \text{Im}(z)$$

Thus the set $C = \{z/z = x + iy, x, y \in R\}$

represents the set of complex numbers.

Two complex numbers

$$z_1 = x_1 + iy_1 \text{ and } z_2 = x_2 + iy_2$$

are equal whenever

$$x_1 = x_2 \text{ and } y_1 = y_2.$$

5.2 Fundamental Operation on Complex Number

Let $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$ be two complex numbers then operations are defined as follows.

i) Addition :

$$\begin{aligned} z_1 + z_2 &= (x_1 + iy_1) + (x_2 + iy_2) \\ &= (x_1 + x_2) + i(y_1 + y_2) \end{aligned}$$

ii) Substraction :

$$\begin{aligned} z_1 - z_2 &= (x_1 + iy_1) - (x_2 + iy_2) \\ &= (x_1 - x_2) + i(y_1 - y_2) \end{aligned}$$

iii) Mutiplication :

$$\begin{aligned} z_1 z_2 &= (x_1 + iy_1)(x_2 + iy_2) \\ &= (x_1 x_2 - y_1 y_2) + i(x_1 y_2 + x_2 y_1) \end{aligned}$$

iv) Division :

$$\begin{aligned} \frac{z_1}{z_2} &= \frac{x_1 + iy_1}{x_2 + iy_2} = \frac{x_1 + iy_1}{x_2 + iy_2} \cdot \frac{x_2 - iy_2}{x_2 - iy_2} \\ &= \left(\frac{x_1 x_2 + y_1 y_2}{x_2^2 + y_2^2} \right) + i \left(\frac{x_2 y_1 - x_1 y_2}{x_2^2 + y_2^2} \right) \end{aligned}$$

provided $z_2 \neq 0$

Ex. 5.2.1 Evaluate

- i) $(4 + 3i) + (-2 - i)$ ii) $(6 - 8i) - (3i - 9)$
 iii) $(3 + 2i)(2 - 3i)$ iv) $(5 - 3i)/(-1 + i)$

Sol. :

$$\begin{aligned} \text{i) } (4 + 3i) + (-2 - i) &= 4 + 3i - 2 - i \\ &= 2 + 2i \end{aligned}$$

$$\begin{aligned} \text{ii) } (6 - 8i) - (3i - 9) &= 6 - 8i - 3i + 9 \\ &= 15 - 11i \end{aligned}$$

$$\begin{aligned} \text{iii) } (3 + 2i)(2 - 3i) &= 3(2 - 3i) + 2i(2 - 3i) \\ &= 6 - 9i + 4i - 6i^2 \\ &= 6 - 9i + 4i + 6 \\ &= 12 - 5i \end{aligned}$$

$$\begin{aligned} \text{iv) } \frac{(5 - 3i)}{-1 + i} &= \frac{5 - 3i}{-1 + i} \times \frac{-1 - i}{-1 - i} \\ &= \frac{-5 - 5i + 3i + 3i^2}{1 - i^2} \\ &= \frac{-8 - 2i}{2} \\ &= -4 - i \end{aligned}$$

5.3 Conjugate of a Complex Number

Conjugate of a complex number $z = x + iy$ is defined as $x - iy$ and is denoted by $\bar{z}$

Two complex numbers z and $\bar{z}$ are said to be complex conjugate of each other.

Properties :

$$\text{i) } \overline{z_1 \pm z_2} = \bar{z}_1 \pm \bar{z}_2 \quad \text{ii) } \overline{z_1 \cdot z_2} = \bar{z}_1 \cdot \bar{z}_2$$

$$\text{iii) } \overline{\left(\frac{z_1}{z_2} \right)} = \frac{\bar{z}_1}{\bar{z}_2} : z_2 \neq 0 \quad \text{iv) } \overline{(\bar{z})} = z$$

$$\text{v) } \frac{z + \bar{z}}{2} = \text{Re}(z) \quad \text{vi) } \frac{z - \bar{z}}{2i} = \text{Im}(z)$$

5.3.1 Modulus of a Complex Number

If $z = x + iy$ is complex number then modulus or absolute value of z is denoted by $|z|$ and is defined by, $|z| = \sqrt{x^2 + y^2}$

Properties of Modulus

- i) $|z_1 \pm z_2| \leq |z_1| + |z_2|$
 ii) $|z_1 \pm z_2| \geq ||z_1| - |z_2||$
 iii) $|z_1 \cdot z_2| = |z_1| \cdot |z_2|$

$$\text{iv) } \left| \frac{z_1}{z_2} \right| = \frac{|z_1|}{|z_2|}$$

$$\text{v) } |z| = |\bar{z}|$$

$$\text{vi) } z \cdot \bar{z} = |z|^2$$

5.4 Graphical Representation of Complex Number

Complex number $x + iy$ can be considered as an ordered pair of real numbers. We can represent such numbers by points in an xy plane called complex plane or Argand diagram.

For example, $1 + 2i$, $-3 - i$, $-2 + i$, $1 - 3i$ are plot in Fig. 5.4.1.

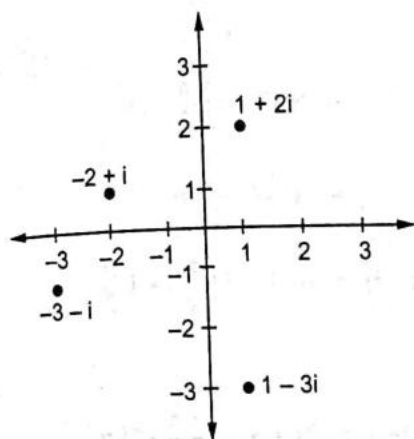


Fig. 5.4.1

The distance between two points $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$ in the complex plane is given by,

$$|z_1 - z_2| = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

5.4.1 Polar Form of Complex Numbers

The polar form of complex number $x + iy$ is represented as,

$$x + iy = r \cos \theta + i r \sin \theta$$

$$= r (\cos \theta + i \sin \theta)$$

where, $r = \sqrt{x^2 + y^2} = |x + iy|$ (modulus or absolute value.)

$$\theta = \tan^{-1} \left(\frac{y}{x} \right) \text{ (Amplitude or principal argument)}$$

Note : Principal argument always lies between

$$-\pi < \theta \leq \pi$$

$$\text{i) If } x > 0, y > 0, \theta = \tan^{-1} \left(\frac{y}{x} \right)$$

$$\text{ii) If } x < 0, y > 0, \theta = \pi - \tan^{-1} \left| \frac{y}{x} \right|$$

$$\text{iii) If } x < 0, y < 0, \theta = -\pi + \tan^{-1} \left(\frac{y}{x} \right)$$

$$\text{iv) If } x > 0, y < 0, \theta = -\tan^{-1} \left| \frac{y}{x} \right|$$

Ex. 5.4.1 Express the following complex numbers in polar form : i) $1 + \sqrt{3}i$, ii) $-2 + 2i$, iii) $-\sqrt{3} - i$, iv) $-2i$

Sol. :

$$\text{i) } 1 + \sqrt{3}i$$

$$r = \sqrt{1+3} = \sqrt{4} = 2$$

$$\theta = \tan^{-1}(\sqrt{3}) = \frac{\pi}{3}$$

$$1 + \sqrt{3}i = 2 \left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right)$$

$$\text{ii) } -2 + 2i$$

$$r = \sqrt{4+4} = \sqrt{8} = 2\sqrt{2}$$

$$\theta = \pi - \tan^{-1}1 = \pi - \frac{\pi}{4} = \frac{3\pi}{4}$$

$$-2 + 2i = 2\sqrt{2} \left(\cos \frac{3\pi}{4} + i \sin \frac{3\pi}{4} \right)$$

$$\text{iii) } -\sqrt{3} - i$$

$$r = \sqrt{3+1} = \sqrt{4} = 2$$

$$\theta = -\pi + \tan^{-1} \left(\frac{1}{\sqrt{3}} \right) = -\pi + \frac{\pi}{6} = -\frac{5\pi}{6}$$

$$-\sqrt{3} - i = 2 \left(\cos \frac{5\pi}{6} - i \sin \frac{5\pi}{6} \right)$$

$$\text{iv) } -2i$$

$$r = \sqrt{0+4} = 2$$

$$\theta = -\pi + \tan^{-1} \left(\frac{2}{0} \right) = -\pi + \frac{\pi}{2} = -\frac{\pi}{2}$$

$$-2i = 2 \left(\cos \frac{\pi}{2} - i \sin \frac{\pi}{2} \right)$$

5.5 Trigonometric Functions of a Complex Variable z

The trigonometric function of complex variable z are defined as,

$$\sin z = \frac{e^{iz} - e^{-iz}}{2i}, \quad \cos z = \frac{e^{iz} + e^{-iz}}{2}$$

$$\tan z = \frac{\sin z}{\cos z}, \quad \cot z = \frac{\cos z}{\sin z}$$

$$\sec z = \frac{1}{\cos z}, \quad \operatorname{cosec} z = \frac{1}{\sin z}$$

5.6 Hyperbolic Functions

The complex hyperbolic functions are defined by,

$$\cosh z = \frac{e^z + e^{-z}}{2}, \quad \sinh z = \frac{e^z - e^{-z}}{2}$$

Note : $\cosh(iz) = \frac{e^{iz} + e^{-iz}}{2} = \cos z$

$$\sinh(iz) = \frac{e^{iz} - e^{-iz}}{2} = i \frac{e^{iz} - e^{-iz}}{2i} = i \sin z$$

The general formulae for the real hyperbolic functions continue to hold for the corresponding complex valued functions.

i) $\sinh(z_1 \pm z_2) = \sinh z_1 \cosh z_2 \pm \cosh z_1 \sinh z_2$

ii) $\cosh(z_1 \pm z_2) = \cosh z_1 \cosh z_2 \pm \sinh z_1 \sinh z_2$

iii) $\cosh^2(z) - \sinh^2(z) = 1$

iv) $\sinh(2z) = 2 \sinh(z) \cosh(z)$

v) $\cosh(2z) = \cosh^2(z) + \sinh^2(z)$

5.7 Logarithm of a Complex Numbers

Let $z = x + iy$

Consider polar co-ordinates

$$x = r \cos \theta, \quad y = r \sin \theta$$

So $r = \sqrt{x^2 + y^2}, \quad \theta = \tan^{-1}\left(\frac{y}{x}\right)$

$$z = x + iy$$

$$= r \cos \theta + i r \sin \theta$$

$$= r (\cos(\theta + 2k\pi) + i \sin(\theta + 2k\pi))$$

$$z = r e^{(\theta + 2k\pi)i}$$

$$\ln z = \ln r + i(\theta + 2k\pi)$$

Which is general value of logarithm and is denoted by $\ln z$

$$\therefore \ln z = \ln r + i(\theta + 2k\pi)$$

If $k = 0$, then

$$\ln z = \ln r + i\theta$$

which is known as principal value of logarithm of z .

Ex. 5.7.1 Find general and principal value of logarithm of the following.

i) i ii) $-i$ iii) $1 + i$ iv) $1 + \sqrt{3}i$

sol. : i) i

$$\ln i = \ln 1 + i\left(\frac{\pi}{2} + 2k\pi\right)$$

$$= i \frac{1}{2}(\pi + 4k\pi)$$

$$\ln(i) = \ln 1 + i\left(\frac{\pi}{2}\right) = i \frac{\pi}{2}$$

ii) $-i$

$$\ln(-i) = \ln 1 + i\left(-\frac{\pi}{2} + 2k\pi\right)$$

$$= i \frac{1}{2}(4k\pi - \pi)$$

$$\ln(i) = \ln 1 + i\left(-\frac{\pi}{2}\right) = -i \frac{\pi}{2}$$

iii) $1 + i$

$$\ln(1 + i) = \ln \sqrt{2} + i\left(\frac{\pi}{4} + 2k\pi\right)$$

$$= \frac{1}{2} \ln 2 + i\left[\frac{1}{4}(8k\pi + \pi)\right]$$

$$\ln(1 + i) = \frac{1}{2} \ln 2 + i \frac{\pi}{4}$$

iv) $1 + \sqrt{3}i$

$$\ln(1 + \sqrt{3}i) = \ln 2 + i\left(\frac{\pi}{3} + 2k\pi\right)$$

$$= \ln 2 + i\left[\frac{1}{3}(6k\pi + \pi)\right]$$

$$\ln(1 + \sqrt{3}i) = \ln 2 + i \frac{\pi}{3}$$

5.8 Function of a Complex Variable

GTU : Summer-10, 19

Let H be a set of complex numbers. A function f defined on H is a rule that assigns to each z in H a complex number w . The number w is called the value of f at z and is denoted by $f(z)$, that is $w = f(z)$. The set H is called the domain at definition of f .

Graphical representation :

We know that $z = x + iy$ can be decomposed into real and imaginary part. We find that $w = f(z)$ can also be decomposed.

i.e. $w = u(x, y) + i v(x, y)$

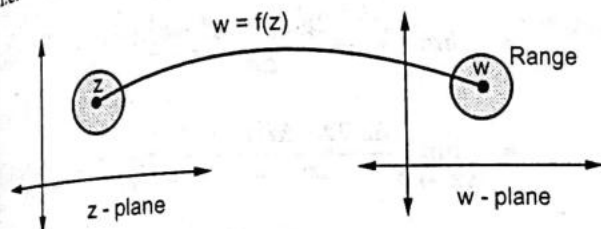


Fig. 5.8.1

Ex. 5.8.1 Write the function $w = z^2 + i$ in the form $u + iv$

Sol. : $w = f(z) = z^2 + i$

$$u + iv = (x + iy)^2 + i \quad (\because z = x + iy)$$

$$= x^2 + 2ixy - y^2 + i$$

$$= (x^2 - y^2) + i(2xy + 1)$$

So, $u(x, y) = x^2 - y^2$ and $v = 2xy + 1$

Ex. 5.8.2 Find the range of the function

$$f(z) = x^2 + i; |z| \leq 1.$$

Sol. : $w = f(z) = u + iv = x^2 + i$

$$u = x^2 \text{ and } v = 1$$

Given that $|z| \leq 1$

$$\Rightarrow x^2 + y^2 \leq 1$$

or $-1 \leq x \leq 1$

$$0 \leq x^2 \leq 1$$

$$\Rightarrow 0 \leq u \leq 1, v = 1$$

$\therefore$ The range is along the line segment from $w = i$ to $w = 1 + i$.

Ex. 5.8.3 Find the value of $\operatorname{Re}(f(z))$ and $\operatorname{Im}(f(z))$ at the indicated point, where.

$$f(z) = \frac{1}{1-z} \text{ at } 7 + 2i$$

Summer - 10, Marks 3

Sol. :

Here $z = x + iy = 7 + 2i$

$$x = 7 \text{ and } y = 2$$

$$= \frac{1}{1-(x+iy)} = \frac{1}{1-(7+2i)} = \frac{1}{1-7-2i}$$

$$= \frac{1}{-6-2i} = \frac{-(6-2i)}{(6+2i)(6-2i)}$$

$$= -\frac{(6-2i)}{36+4} = -\frac{6}{40} + \frac{1}{20}i$$

$$\operatorname{Re} f(z) = -\frac{3}{20}, \quad \operatorname{Im}(f(z)) = \frac{1}{20}$$

Ex. 5.8.4 Write the function $f(z) = z^2 + z + 1$ in the form $f(z) = u(x, y) + iv(x, y)$

Sol. :

$$u + iv = (x + iy)^2 + (x + iy) + 1$$

$$= x^2 + 2ixy - y^2 + x + iy + 1$$

$$= (x^2 - y^2 + x + 1) + i(2xy + y)$$

$$u = x^2 - y^2 + x + 1, v = 2xy + y$$

Ex. 5.8.5 Find real and imaginary part of $f(z) = z^2 + 4z$. Also, calculate the value of f at $z = 1 + i$.

GTU : Summer-19, Maths-4, Marks 4

Sol. :

$$f(z) = z^2 + 4z$$

$$u + iv = (x + iy)^2 + 4(x + iy)$$

$$= x^2 + 2ixy - y^2 + 4x + 4iy$$

$$= (x^2 - y^2 + 4x) + i(2xy + 4y)$$

$$\therefore u = x^2 - y^2 + 4x \text{ (Real part)}$$

$$\text{and } v = 2xy + 4y \text{ (Imaginary part)}$$

$$f(1 + i) = (1 + i)^2 + 4(1 + i)$$

$$= 1 + 2i - 1 + 4 + 4i$$

$$= 4 + 6i$$

5.9 Polar Form of a Complex Function

The Cartesian of a complex function is

$$f(z) = u_{(x,y)} + iv_{(x,y)} \text{ for } z = x + iy$$

$$\operatorname{Re}(f(z)) = u_{(x,y)} \text{ and } \operatorname{Im}(f(z)) = v_{(x,y)}$$

Now we put, $x = r \cos \theta$, $y = r \sin \theta$

then

$$z = x + iy = r \cos \theta + ir \sin \theta$$

$$z = re^{i\theta} (\because \cos \theta + i \sin \theta = e^{i\theta})$$

with

$$|z| = r$$

and

$$\theta = \arg(z);$$

Thus the polar form at a complex function is

$$f(z) = u(r, \theta) + iv(r, \theta)$$

Ex. 5.9.1 Write the function $f(z) = z^2 + z + 2$ in the polar form.

Sol. : $f(z) = z^2 + z + 2$

$$u + iv = (r \cdot e^{i\theta})^2 + r \cdot e^{i\theta} + 2$$

$$= r^2 e^{2i\theta} + r \cdot e^{i\theta} + 2$$

$$= r^2(\cos 2\theta + i \sin 2\theta) + r(\cos \theta + i \sin \theta) + 2$$

$$u + iv = (r^2 \cos 2\theta + r \cos \theta + 2) + i(r^2 \sin 2\theta + r \sin \theta)$$

$$u = r^2 \cos 2\theta + r \cos \theta + 2 \text{ and}$$

$$v = r^2 \sin 2\theta + r \sin \theta$$

Ex. 5.9.2 Find the image in w -plane under the transformation $w = z + 1$ where region at z -plane is $|z| = 1$.

Sol. : $|z| = 1 \Rightarrow \text{radius} = 1, \text{Center } (0, 0)$

$$w = z + 1 \Rightarrow z = w - 1$$

$$|w - 1| = 1$$

$$|u + iv - 1| = 1$$

$$|(u - 1) + iv| = 1$$

Center (1, 0) radius 1 in w - plane.

5.10 Complex Derivative

GTU : Summer-10

Let $w = f(z)$ be a complex function of z defined in a domain D . Then $f(z)$ is said to be differentiable at $z = z_0$ if the limit

$$f'(z) = \lim_{z \rightarrow z_0} \frac{f(z) - f(z_0)}{z - z_0} = \lim_{\Delta z \rightarrow 0} \frac{f(z + \Delta z) - f(z)}{\Delta z}$$

Ex. 5.10.1 By definition P.T. $f(z) = z^2$ is differentiable for all value of z .

Sol. :

$$\begin{aligned} f'(z) &= \lim_{\Delta z \rightarrow 0} \frac{f(z + \Delta z) - f(z)}{\Delta z} = \lim_{\Delta z \rightarrow 0} \frac{(z + \Delta z)^2 - z^2}{\Delta z} \\ &= \lim_{\Delta z \rightarrow 0} \frac{z^2 + 2z\Delta z + \Delta z^2 - z^2}{\Delta z} \\ &= \lim_{\Delta z \rightarrow 0} \frac{\Delta z(2z + \Delta z)}{\Delta z} \end{aligned}$$

$$f'(z) = 2z$$

So $f(z)$ is differentiable for all value of z .

Ex. 5.10.2 Find the value of the derivative of $\frac{z-i}{z+i}$ at i .

Summer-10, Marks 3

Sol. : $f(z) = \frac{z-i}{z+i}$

$$f'(z) = \frac{1(z+i) - 1(z-i)}{(z+i)^2} = \frac{z+i-z+i}{(z+i)^2}$$

$$f'(z)|_{z=i} = \frac{2i}{(i+i)^2} = \frac{1}{2i}$$

Ex. 5.10.3 P.T. $f(z) = \bar{z}$ is not differentiable at any point in the complex plane.

Sol. : $f'(z) = \lim_{\Delta z \rightarrow 0} \frac{f(z + \Delta z) - f(z)}{\Delta z}$

$$= \lim_{\Delta z \rightarrow 0} \frac{\overline{z + \Delta z} - \bar{z}}{\Delta z}$$

$$= \lim_{\Delta z \rightarrow 0} \frac{\bar{z} + \Delta \bar{z} - \bar{z}}{\Delta z} = \lim_{\Delta x \rightarrow 0} \frac{\Delta x - i \Delta y}{\Delta x + i \Delta y} \quad \Delta y \rightarrow 0$$

i) Along x-axis, $\Delta y \rightarrow 0$

$$= \lim_{\Delta x \rightarrow 0} \frac{\Delta x}{\Delta x} = 1$$

ii) Along y-axis, $\Delta x \rightarrow 0$

$$= \lim_{\Delta y \rightarrow 0} \frac{-i \Delta y}{i \Delta y} = -1$$

So limit does not exist. So $f'(z) = \bar{z}$ is not differentiable at any point of z .

Ex. 5.10.4 By definition P.T. $f(z) = \frac{1}{z}$ is not differentiable

at $z = 0$.

$$\text{Sol. : } f'(z) = \lim_{\Delta z \rightarrow 0} \frac{f(z + \Delta z) - f(z)}{\Delta z}$$

$$= \lim_{\Delta z \rightarrow 0} \frac{\frac{1}{z + \Delta z} - \frac{1}{z}}{\Delta z} = \lim_{\Delta z \rightarrow 0} \frac{z - z - \Delta z}{\Delta z z(z + \Delta z)}$$

$$= \lim_{\Delta z \rightarrow 0} \frac{-\Delta z}{\Delta z z(z + \Delta z)} = -\frac{1}{z^2}$$

So $f'(z)$ is not differentiable at $z = 0$.

5.11 Analytic Functions

GTU : Winter-12,18,19,20,21,22,23,24,
Summer-10,18,19,20,21,22,23, Marks 7

A single valued function $f(z)$ is said to be analytic at the point z_0 in the domain D of the z -plane if $f(z)$ is differentiable at $z = z_0$ and also differentiable its neighbourhood at z_0 .

An analytic function is also known as holomorphic or regular or monogenic.

Entire function : A function which is analytic any where in the complex plane is said to be entire function.

Theorem : The necessary conditions for $f(z)$ to be analytic.

State : If $f(z) = u_{(x,y)} + i v_{(x,y)}$ is differentiable at any point $z = x + iy$ then the real and imaginary parts of $u(x, y)$ and $v(x, y)$ of $f(z)$ are also differentiable at

(x, y) and the C - R equation $y_x = v_y$ and $u_y = -v_x$ is satisfied.

Proof : Here given that $f(z)$ is differentiable.

So $f'(z) = \lim_{\Delta z \rightarrow 0} \frac{f(z + \Delta z) - f(z)}{\Delta z}$ is satisfied.

$$f'(z) = \lim_{\Delta z \rightarrow 0} \frac{f(z + \Delta z) - f(z)}{\Delta z}$$

$$f'(z) = \lim_{\substack{\Delta x \rightarrow 0 \\ \Delta y \rightarrow 0}} \frac{u_{(x + \Delta x, y + \Delta y)} + i v_{(x + \Delta x, y + \Delta y)} - u_{(x, y)} - i v_{(x, y)}}{\Delta x + i \Delta y}$$

$$(\because f(z) = u_{(x, y)} + i v_{(x, y)})$$

$$f(z + \Delta z) = u_{(x + \Delta x, y + \Delta y)} + i v_{(x + \Delta x, y + \Delta y)}$$

Now, we can take two cases.

1) Along x-axis, $\Delta y \rightarrow 0$.

$$f'(z) = \lim_{\Delta x \rightarrow 0} \frac{u_{(x + \Delta x, y)} + i v_{(x + \Delta x, y)} - u_{(x, y)} - i v_{(x, y)}}{\Delta x}$$

$$= \lim_{\Delta x \rightarrow 0} \frac{u_{(x + \Delta x, y)} - u_{(x, y)}}{\Delta x} + i \lim_{\Delta x \rightarrow 0} \frac{v_{(x + \Delta x, y)} - v_{(x, y)}}{\Delta x}$$

$$f'(z) = u_x + i v_x \quad (\text{By definition}) \quad \dots (A)$$

2) Along y - axis, $\Delta x \rightarrow 0$

$$\begin{aligned} \text{So } f'(z) &= \lim_{\Delta y \rightarrow 0} \frac{u_{(x, y + \Delta y)} - u_{(x, y)}}{i \Delta y} \\ &\quad + i \lim_{\Delta y \rightarrow 0} \frac{v_{(x, y + \Delta y)} - v_{(x, y)}}{i \Delta y} \\ &= -i u_y + v_y \quad (\text{By definition}) \quad \dots (B) \end{aligned}$$

From equations (A) and (B)

$$u_x = v_y \quad \text{and} \quad u_y = -v_x$$

(Compare real and imaginary part)

Sufficient condition for analytic function

State : If u and v of $f(z)$ are both differentiable at (x, y) and satisfy the C-R equation $u_x = v_y$ and $u_y = -v_x$ then $f(z)$ is differentiable at $z = x + iy$.

Note 1 : C-R equation are only necessary but not sufficient for a functions to be differentiable at a point.

Note 2 : If $f(z)$ is analytic every where in domain D then $f(z)$ is differentiable every where in D and also C-R equation satisfied in D .

Note 3 : If $f(z)$ is analytic at only $z = z_0$ then $f(z)$ is differentiable only at $z = z_0$ and C-R equation satisfied only at $z = z_0$.

Note 4 : If $f(z)$ is differentiable at $z = z_0$ then it is not analytic at $z = z_0$ but C-R equation satisfied at $z = z_0$.

Note 5 : If C-R equation is not satisfied at z_0 then $f(z)$ is not differentiable at $z = z_0$ and not analytic at $z = z_0$.

Note 6 : If $f(z)$ and $g(z)$ are analytic then $f \pm g$, $f \cdot g$ is analytic and f/g are analytic if $g(z) \neq 0$.

Note 7 : If $f(z)$ is analytic then it is continuous.

Ex. 5.11.1 Give one example which is not analytic at $z = 0$ but C-R equation is satisfied at $z = 0$. or Show that

$$f(z) = \begin{cases} \sqrt{|xy|}; & z \neq 0 \\ 0; & z = 0 \end{cases}$$

is not analytic at $z = 0$ but C-R equation is satisfied at $z = 0$.

Sol. : Here $u = \sqrt{|xy|}$ $v = 0$

$$\frac{\partial u}{\partial x} = \lim_{\Delta x \rightarrow 0} \frac{u(x + \Delta x, y) - u(x, y)}{\Delta x}$$

$$\left. \frac{\partial u}{\partial x} \right|_{(0,0)} = \lim_{\Delta x \rightarrow 0} \frac{u(\Delta x, 0) - u(0, 0)}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{0 - 0}{\Delta x} = 0$$

$$\left. \frac{\partial u}{\partial y} \right|_{(0,0)} = \lim_{\Delta y \rightarrow 0} \frac{v(0, \Delta y) - v(0, 0)}{\Delta y} = 0$$

$$u_x = v_y \quad \text{and} \quad u_y = -v_x \quad \text{at } (0, 0)$$

So C - R equation satisfied at $z = 0$.

Now, we prove that $f(z)$ is not different at $z = 0$.

$$\begin{aligned} f'(z) &= \lim_{z \rightarrow 0} \frac{f(z) - f(0)}{z - 0} = \lim_{z \rightarrow 0} \frac{\sqrt{|xy|} - 0}{x + iy} \\ &= \lim_{\substack{\Delta x \rightarrow 0 \\ \Delta y \rightarrow 0}} \frac{\sqrt{|xy|}}{x + iy} \end{aligned}$$

Along x-axis, $y \rightarrow 0$

$$\lim_{x \rightarrow 0} \frac{0}{x} = 0$$

Along y-axis, $x \rightarrow 0$

$$\lim_{y \rightarrow 0} \frac{0}{iy} = 0$$

Along $y = mx$

$$\lim_{x \rightarrow 0} \frac{\sqrt{x \cdot mx}}{x + i \cdot mx} = \lim_{x \rightarrow 0} \frac{x \sqrt{m}}{x(1 + im)} = \frac{\sqrt{m}}{1 + im}$$

So limit does not exist.

So $f(z)$ is not different at $z = 0$.

Ex. 5.11.2 Show that

$$f(z) = \begin{cases} \frac{\bar{z}^2}{z} & z \neq 0 \\ 0 & z = 0 \end{cases}$$

C-R equation satisfied at $z = 0$.

GTU : Winter-18, Summer-20, CVNM, Marks 7

Sol. : $f(z) = \frac{\bar{z}^2}{z} = \frac{(x - iy)^2}{x + iy}$

$$= \frac{(x^2 - 2ixy - y^2)(x - iy)}{(x + iy)(x - iy)}$$

$$= \frac{(x^2 - y^2 - 2ixy)(x - iy)}{x^2 + y^2}$$

$$= \frac{x^3 - x^2iy - y^2x + iy^3 - 2ix^2y - 2xy^2}{x^2 + y^2}$$

$$= \frac{x^3 - 3xy^2}{x^2 + y^2} + i \frac{(y^3 - 3x^2y)}{x^2 + y^2}$$

$$u = \frac{x^3 - 3xy^2}{x^2 + y^2} \quad v = \frac{y^3 - 3x^2y}{x^2 + y^2}$$

So

By definition

$$u_x(0, 0) = \lim_{\Delta x \rightarrow 0} \frac{u(0+\Delta x, 0) - u(0, 0)}{\Delta x}$$

$$= \lim_{\Delta x \rightarrow 0} \frac{\frac{\Delta x^3}{\Delta x^2} - 0}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{\Delta x^3}{\Delta x^3} = 1$$

$$u_y(0, 0) = \lim_{\Delta y \rightarrow 0} \frac{u(0, 0+\Delta y) - u(0, 0)}{\Delta y} = \lim_{\Delta y \rightarrow 0} \frac{0}{\Delta y} = 0$$

$$v_x(0, 0) = \lim_{\Delta x \rightarrow 0} \frac{v(\Delta x, 0) - v(0, 0)}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{0-0}{\Delta x} = 0$$

$$v_y(0, 0) = \lim_{\Delta y \rightarrow 0} \frac{v(0, \Delta y) - v(0, 0)}{\Delta y} = \lim_{\Delta y \rightarrow 0} \frac{\Delta y}{\Delta y} = 1$$

So, $u_x = v_y$ and $u_y = -v_x$ Satisfied at $z = 0$.

Ex. 5.11.3 Examine the nature of function.

$$f(z) = \frac{x^2 y^5 (x+iy)}{x^4 + y^{10}}, \quad z \neq 0$$

$$0, \quad z = 0$$

in the region including the origin.

$$\text{Sol. : } f(z) = u + iv = \frac{x^3 y^5}{x^4 + y^{10}} + i \frac{x^2 y^6}{x^4 + y^{10}}$$

$$u = \frac{x^3 y^5}{x^4 + y^{10}}, \quad v = \frac{x^2 y^6}{x^4 + y^{10}}$$

$$u_x(0, 0) = \lim_{\Delta x \rightarrow 0} \frac{u(\Delta x, 0) - u(0, 0)}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{0-0}{\Delta x} = 0$$

$$u_y(0, 0) = \lim_{\Delta y \rightarrow 0} \frac{u(0, \Delta y) - u(0, 0)}{\Delta y} = \lim_{\Delta y \rightarrow 0} \frac{0-0}{\Delta y} = 0$$

$$v_x(0, 0) = \lim_{\Delta x \rightarrow 0} \frac{v(\Delta x, 0) - v(0, 0)}{\Delta x} = \lim_{\Delta x \rightarrow 0} \frac{0-0}{\Delta x} = 0$$

$$v_y(0, 0) = \lim_{\Delta y \rightarrow 0} \frac{v(0, \Delta y) - v(0, 0)}{\Delta y} = \lim_{\Delta y \rightarrow 0} \frac{0-0}{\Delta y} = 0$$

Hence C - R equation satisfied at origin.

Now we check $f(z)$ is not differential at $z = 0$.

$$f'(0) = \lim_{z \rightarrow 0} \frac{f(z) - f(0)}{z - 0} = \lim_{z \rightarrow 0} \frac{\left[\frac{x^2 y^5 (x+iy)}{x^4 + y^{10}} - 0 \right]}{x+iy}$$

$$= \lim_{\substack{x \rightarrow 0 \\ y \rightarrow 0}} \frac{x^2 y^5}{x^4 + y^{10}}$$

i) Along $y = mx$

$$= \lim_{x \rightarrow 0} \frac{x^2 m^5 x^5}{x^4 + m^{10} x^{10}} = \lim_{x \rightarrow 0} \frac{x^3 m^5}{1 + m^{10} x^6} = 0$$

ii) Along $y^5 = x^2$

$$= \lim_{x \rightarrow 0} \frac{x^2 x^2}{x^4 + x^4} = \frac{1}{2}$$

So limit does not exist.

So function is not differential at $z = 0$.Ex. 5.11.4 Verify Cauchy - Rieman equation for $f(z) = \cos x \cdot \cosh y - i \sin x \cdot \sinh y$.

GTU : Summer-20, CVNM, Marks 3

$$\text{Sol. : } f(z) = \cos x \cdot \cosh y - i \sin x \cdot \sinh y$$

$$u + iv = \cos x \cdot \cosh y - i \sin x \cdot \sinh y$$

$$u = \cos x \cdot \cosh y \quad v = -\sin x \cdot \sinh y$$

$$y_x = -\sin x \cdot \cosh y \quad v_x = -\cos x \cdot \sinh y$$

$$u_y = \cos x \cdot \sinh y \quad v_y = -\sin x \cdot \cosh y$$

$$\therefore \text{C-R equation : } y_x = v_y \text{ and } u_y = -v_x$$

 $\therefore$ C-R equation is satisfied.Ex. 5.11.5 Show that $f(z) = e^{\bar{z}}$ is nowhere analytic.

GTU : Summer-22, CVNM, Marks 4

$$\text{Sol. : } f(z) = e^{\bar{z}}$$

$$u + iv = e^{x-iy}$$

$$= e^x (\cos y - i \sin y)$$

$$= e^x \cos y - i e^x \cdot \sin y$$

$$\therefore u = e^x \cdot \cos y \quad v = -e^x \cdot \sin y$$

$$u_x = e^x \cdot \cos y \quad v_x = -e^x \cdot \sin y$$

$$u_y = -e^x \cdot \sin y \quad v_y = -e^x \cdot \cos y$$

$\therefore$ C-R equation : $u_x \neq v_y$ and $u_y \neq -v_x$

$\therefore$ C-R equation is not satisfied.

$\therefore f(z)$ is not analytic.

Ex. 5.11.6 Which is the following function are entire.

i) $f(z) = 3x + y + i(3y - x)$

ii) $f(z) = xy + iy$ iii) $f(z) = e^y \cdot e^{ix}$

iv) $f(z) = \bar{z} + 1$ v) $f(z) = (z^2 - 1)e^{-x} \cdot e^{-iy}$

vi) $f(z) = z^2 + 2z + 1$ vii) $f(z) = \bar{z}$

**GTU : Summer-10,19,
Winter-19, 20, CVNM, Marks 3,
Summer-23, Marks 4**

Sol. :

i) $f(z) = (3x + y) + i(3y - x) = u + iv$

$$u = 3x + y \quad v = 3y - x$$

$$u_x = 3 \quad v_x = -1$$

$$u_y = 1 \quad v_y = 3$$

So C - R equation is satisfied every where.

u_x, u_y, v_x and v_y continuous every where.

So $f(z)$ is differentiable every where.

So $f(z)$ is analytic every where.

So $f(z)$ is entire function.

ii) $f(z) = xy + iy$

$$u = xy \quad v = y$$

$$u_x = y \quad u_y = x$$

$$v_x = 0 \quad v_y = 1$$

So $u_x = v_y \quad u_y = -v_x$

$$y = 1 \quad x = 0$$

So C - R equation satisfied only $x = 0, y = 1$ means $z = i$.

u_x, u_y, v_x and v_y continuous every where.

So $f(z)$ is differentiable only $z = i$.

So no where analytic.

So $f(z)$ is not entire function.

iii) $f(z) = e^y \cdot e^{ix}$

$$u + iv = e^y (\cos x + i \sin x) = e^y \cos x + i e^y \sin x$$

$$u = e^y \cos x \quad v = e^y \sin x$$

$$u_x = -e^y \sin x \quad v_x = e^y \cos x$$

$$u_y = e^y \cos x \quad v_y = e^y \sin x$$

So C - R equation is not satisfied any where.

So $f(z)$ is no where differentiable.

So $f(z)$ is no where analytic.

So $f(z)$ is not entire function.

iv) $f(z) = \bar{z} + 1$

$$u + iv = x - iy + 1 = (x + 1) - iy$$

$$u = x + 1 \quad v = -y$$

$$u_x = 1 \quad v_x = 0$$

$$u_y = 1 \quad v_y = -1$$

So C - R equation is not satisfied in z - plane.

So $f(z)$ is no where analytic.

So $f(z)$ is not entire function.

v) $f(z) = (z^2 - 1)e^{-z}$

Here $f(z) = g(z) h(z)$

$$g(z) = z^2 - 1 \text{ is analytic any where.}$$

$$h(z) = e^{-z} \text{ is also analytic any where.}$$

So $f(z)$ is analytic any where.

So $f(z)$ is entire function.

vi) $f(z) = z^2 + 2z + 1$

Here $f(z) = z^2 + 2z + 1$ is analytic any where because we know that any polynomial function is analytic.

So $f(z)$ is entire function.

vii) $f(z) = \bar{z}$
 $u + iv = x - iy$
 $u = x \quad v = -y$
 $u_x = 1 \quad v_x = 0$
 $u_y = 0 \quad v_y = -1$

So C - R equation is not satisfied.

So $f(z) = \bar{z}$ is no where analytic.

So $f(z)$ is not entire function.

Ex. 5.11.7 Define entire function and show that $f(z) = e^z$ is entire function.

GTU : Winter-18, Maths-4, Marks 3
Summer-18, CVNM, Marks 3

Sol. : Entire function : A function which is analytic every where in the complex plane is said to be entire function.

$$f(z) = e^z$$

$$u + iv = e^{x+iy}$$

$$u + iv = e^x(\cos y + i \sin y)$$

$$u = e^x \cos y$$

$$v = e^x \cdot \sin y$$

$$u_x = e^x \cdot \cos y$$

$$v_x = e^x \cdot \sin y$$

$$u_y = -e^x \cdot \sin y$$

$$v_y = e^x \cdot \cos y$$

∴ C-R equation : $u_x = v_y$ and $u_y = -v_x$

∴ C-R equation satisfied every where u_x, u_y, v_x and v_y continuous every where

∴ $f(z)$ is analytic every where

∴ $f(z)$ is entire.

Ex. 5.11.8 $f(z) = |z|^2$ is analytic ?

GTU : Winter-22, Maths-4, Marks 3

Sol. : $f(z) = x^2 + y^2$

$$u = x^2 + y^2 \quad v = 0$$

$$u_x = 2x, \quad u_y = 2y, \quad v_x = 0 \quad v_y = 0$$

$$u_x = v_y \quad \text{and} \quad u_y = -v_x$$

$$2x = 0 \quad 2y = 0$$

$$x = 0 \quad \text{and} \quad y = 0$$

So, C - R equation satisfied only $x = 0, y = 0, (z = 0)$.

u_x, u_y, v_x and v_y continuous any where

So $f(z)$ is differentiable only $z = 0$.

So $f(z)$ is no where analytic.

Ex. 5.11.9 Prove that $f(z) = \frac{1}{z}$ is differentiable every where, except $z = 0$.

Sol. :

$$u + iv = \frac{1}{x+iy} = \frac{x-iy}{(x+iy)(x-iy)} = \frac{x-iy}{x^2+y^2}$$

$$= \frac{x}{x^2+y^2} - i \frac{y}{x^2+y^2}$$

$$u = \frac{x}{x^2+y^2} \quad v = \frac{-y}{x^2+y^2}$$

$$u_x = \frac{1 \cdot (x^2+y^2) - x(2x)}{(x^2+y^2)^2} = \frac{y^2+x^2}{(x^2+y^2)^2}$$

$$u_y = 0 - \frac{2xy}{(x^2+y^2)^2} = \frac{-2xy}{(x^2+y^2)^2}$$

$$v_x = 0 + \frac{2xy}{(x^2+y^2)^2}, \quad v_y = \frac{y^2-x^2}{(x^2+y^2)^2}$$

So, C - R equation satisfied any where.

But, u_x, u_y, v_x and v_y continuous any where except $x = 0, y = 0$ (means $z = 0$)

So $f(z)$ is differential any where except $z = 0$.

So $f(z)$ is analytic any where except $z = 0$.

But, $f(z)$ is not entire.

Ex. 5.11.10 Define analytic function. Check analyticity of $f(z) = 2x + iy^2x$.

GTU : Summer-18, Maths-4, Marks 4

Sol. : Analytic function : A single valued function $f(z)$ is said to be analytic at the point z_0 in the domain D of the z -plane if $f(z)$ is differentiable at $z = z_0$ and also differentiable its neighborhood at z_0 .

Here, $f(z) = 2x + ixy^2$

$$\begin{aligned}\therefore u + iv &= 2x + ixy^2 \\ \therefore u &= 2x, \quad v = xy^2 \\ \therefore u_x &= 2, \quad v_x = y^2 \\ u_y &= 0, \quad v_y = 2xy\end{aligned}$$

$\therefore$ C-R equation

$$\begin{aligned}u_x &= v_y, \quad u_y = -v_x \\ 2 &= 2xy, \quad 0 = -y^2\end{aligned}$$

- $\therefore 2 \neq 0$ (Put $y = 0$) $\therefore y = 0$
 $\therefore$ C-R equation is not satisfied.
 $\therefore$ Given function is not analytic.

Ex. 5.11.11 State necessary and sufficient conditions for the function to be analytic. Also show that $f(z) = |z|$ is not an analytic function.

GTU : Winter-22, Maths-4, Marks 7

Sol. :

- Necessary condition :** If $f(z) = u + iv$ is differentiable at any point $z = x + iy$ then the real and imaginary parts of $u(x, y)$ and $v(x, y)$ of $f(z)$ are also differentiable at (x, y) and the C-R equation. $u_x = v_y$ and $u_y = -v_x$ is satisfied.
- Sufficient condition :** If u and v of $f(z)$ are both differentiable of (x, y) and satisfy the C-R equation $u_x = v_y$ and $u_y = -v_x$, then $f(z)$ is differentiable at $z = x + iy$.
- Now we prove that $f(z) = |z|$ is not analytic function.

$$\begin{aligned}f(z) &= |z| \\ u + iv &= \sqrt{x^2 + y^2} \\ \therefore u &= \sqrt{x^2 + y^2} \quad v = 0 \\ u_x &= \frac{1 \cdot 2x}{2\sqrt{x^2 + y^2}} \quad v_x = 0 \\ \therefore u_x &= \frac{x}{\sqrt{x^2 + y^2}} \quad v_x = 0 \\ u_y &= \frac{y}{\sqrt{x^2 + y^2}} \quad v_y = 0\end{aligned}$$

- $\therefore$ C-R equation is not satisfied.
 $\therefore f(z)$ is not analytic function.

Ex. 5.11.12 Check whether the following functions are analytic or not at any point
 i) $f(z) = x^2 + ixy$ ii) $f(z) = z^2$

GTU : Summer-19, Maths-4, Marks 4
Summer-22, Marks 4
Winter-22, CVNM, Marks 7

Sol. :

i) $f(z) = x^2 + ixy$

$$u + iv = x^2 + ixy$$

$$u = x^2 \quad \text{and} \quad v = xy$$

$$u_x = 2x \quad v_x = y$$

$$u_y = 0 \quad v_y = x$$

Here, $u_x \neq v_y$ and $u_y \neq -v_x$

$\therefore$ C-R equation is not satisfied.

$\therefore f(z)$ is not analytic.

ii) $f(z) = z^2$

$$u + iv = (x + iy)^2$$

$$u + iv = x^2 + 2ixy - y^2$$

$$u = x^2 - y^2 \quad v = 2xy$$

$$u_x = 2x \quad v_x = 2y$$

$$u_y = -2y \quad v_y = 2x$$

$$u_x = v_y \quad \text{and} \quad u_y = -v_x$$

$\therefore$ C-R equation is satisfied every where.

$\therefore u_x, u_y, v_x$ and v_y continuous every where.

$\therefore f(z)$ is analytic every where.

Ex. 5.11.13 Using the C-R equation, show that $f(z) = z^3$ is analytic every where.

GTU : Winter-20, Maths-4, Marks 3
Summer-21, CVNM, Marks 4
Winter-24, CVPDE, Marks 3

Sol. :

$$f(z) = z^3$$

$$u + iv = (x + iy)^3$$

$$= x^3 + (iy)^3 + 3x^2(iy) + 3x(iy)^2$$

$$= x^3 - iy^3 + i3x^2y - 3xy^2$$

$$u + iv = (x^3 - 3xy^2) + i(3x^2y - y^3)$$

$$u = x^3 - 3xy^2 \quad v = 3x^2y - y^3$$

$$u_x = 3x^2 - 3y^2 \quad v_x = 6xy$$

$$u_y = -6xy \quad v_y = 3x^2 - 3y^2$$

$\therefore$ C-R equation satisfied every where u_x, u_y, v_x and v_y continuous every where.

$\therefore f(z)$ is analytic every where.

Ex. 5.11.14 Show that $f(z) = \sin z$ is analytic every where.

GTU : Winter-18, Summer-19, CVNM, Marks 4
Summer-23, CVNM, Marks 3
Winter-24, CVNM, Marks 7

$$f(z) = \sin z$$

$$u + iv = \sin(x + iy)$$

$$= \sin x \cdot \cos iy + \cos x \cdot \sin iy$$

$$\cos iy = \cosh y \text{ and } \sin iy = i \sinh y$$

$$u + iv = \sin x \cdot \cosh y + i \cos x \cdot \sinh y$$

$$u = \sin x \cdot \cosh y \quad v = \cos x \cdot \sinh y$$

$$u_x = \cos x \cdot \cosh y \quad v_x = -\sin x \cdot \sinh y$$

$$u_y = \sin x \cdot \sinh y \quad v_y = \cos x \cdot \cosh y$$

$\therefore$ C-R equation $u_x = v_y$ and $u_y = -v_x$

$\therefore$ C-R equation is satisfied every where.

u_x, u_y, v_x and v_y is continuous every where.

$\therefore f(z)$ is analytic every where.

Ex. 5.11.15 If $u - v = (x - y)(x^2 + 4xy + y^2)$ and $f(z) = u + iv$ is an analytic function at $z = x + iy$

Find $f(z)$ in terms of z . **GTU : Winter-12, Marks 4**

Sol. : $(1 + i)f(z) = (u - v) + i(u + v)$

$$F(z) = U + iV$$

$$u - v = U = (x - y)(x^2 + 4xy + y^2)$$

$$U_x = 1(x^2 + 4xy + y^2) + (x - y)(2x + 4y)$$

$$U_y = (-1)(x^2 + 4xy + y^2) + (x - y)(4x + 2y)$$

$$U_{x(z,0)} = (z^2) + (z - 0)(2z) = z^2 + 2z^2 = 3z^2$$

$$3z^2 \quad U_{y(z,0)} = -(z^2) + (z - 0)(4z) = -z^2 + 4z^2 =$$

$$F(z) = \int U_{x(z,0)} dz - i \int U_{y(z,0)} dz$$

$$= \int 3z^2 dz - i \int 3z^2 dz = z^3 - iz^3 + C$$

$$(1 + i)f(z) = z^3(1 - i) + C$$

$$f(z) = z^3 \frac{(1 - i)}{(1 + i)} + \frac{C}{1 + i} = -iz^3 + C_1$$

Ex. 5.11.16 If $f(z) = u + iv$ is an analytic function of

$$z = x + iy \text{ and } u - v = \frac{e^y - \cos x + \sin x}{\cosh y - \cos x}$$

Sol. :

$$(1 + i)f(z) = (u - v) + i(u + v)$$

$$F(z) = U + iV$$

$$(u - v) = U = \frac{e^y - \cos x + \sin x}{\cosh y - \cos x}$$

$$U_x = +(\sin x + \cos x)(\cosh y - \cos x) - \frac{(e^y - \cos x + \sin x)(+\sin x)}{(\cosh y - \cos x)^2}$$

$$U_{x(z,0)} = \frac{(\sin z + \cos z)(1 - \cos z) - (1 - \cos z + \sin z)(\sin z)}{(1 - \cos z)^2}$$

$$= \frac{\sin z - \sin z \cos z + \cos z - \cos^2 z - \sin z + \sin z \cos z - \sin^2 z}{(1 - \cos z)^2}$$

$$= \frac{\cos z - (\cos^2 z + \sin^2 z)}{(1 - \cos z)^2}$$

$$= \frac{\cos z - 1}{(1 - \cos z)^2} = \frac{1}{\cos z - 1}$$

$$U_y = \frac{e^y(\cosh y - \cos x) - (e^y - \cos x + \sin x)(\sinh y)}{(\cosh y - \cos x)^2}$$

$$U_{y(z,0)} = \frac{e^0(1 - \cos z) - (e^z - \cos z + \sin z)(0)}{(1 - \cos z)^2}$$

$$= \frac{1}{1 - \cos z}$$

$$F(z) = \int U_{x(z,0)} dz - i \int U_{y(z,0)} dz$$

$$= \int \frac{1}{\cos z - 1} dz - i \int \frac{1}{1 - \cos z} dz$$

$$= - \int \frac{1}{2 \sin^2 z/2} dz - i \int \frac{1}{2 \sin^2 z/2} dz$$

$$= -\frac{1}{2} \int \operatorname{cosec}^2 z/2 dz - \frac{i}{2} \int \operatorname{cosec}^2 z/2 dz$$

$$= -\frac{1}{2} \frac{\cot(z/2)}{1/2} - \frac{i}{2} \frac{\cot z/2}{1/2} + C$$

$$= -(1+i) \cot(z/2) + C$$

$$(1+i) f(z) = -(1+i) \cot(z/2) + C$$

$$f(z) = -\cot(z/2) + \frac{C}{(1+i)}$$

$$f(z) = \cot(z/2) + C_1$$

Ex. 5.11.17 If $f(z) = u + iv$ is an analytic function of z and $u + v = x^2 - y^2 + 2xy$, find $f(z)$.

GTU : Summer-24, Maths-4, Marks 7

Sol. : $f(z) = u + iv$

$$i f(z) = iu - v$$

$$\therefore (1+i) f(z) = (u - v) + i(u + v)$$

$$\therefore F(z) = U + iV$$

Where, $F(z) = (1+i) f(z)$, $U = u - v$, $V = u + v$

$$\therefore V = (u + v) = x^2 - y^2 + 2xy$$

By Milne Thomson's method,

$$F(z) = \int V_y(z,0) dz + i \int V_x(z,0) dz + V(0,0)$$

$$V(0,0) = 0$$

$$V_x = 2x + 2y \Rightarrow V_x(z,0) = 2z$$

$$V_y = -2y + 2x \Rightarrow V_y(z,0) = 2z$$

$$\therefore F(z) = \int 2z dz + i \int 2z dz$$

$$= z^2 + iz^2 + C$$

$$(1+i) f(z) = z^2(1+i) + C$$

$$\therefore f(z) = z^2 + \frac{C}{1+i}$$

Ex. 5.11.18 If $f(z) = u + iv$ and $u - v = e^x(\cos y - \sin y)$, find $f(z)$ in terms of z .

GTU : Summer-21, Winter-21, Maths-4, Marks 7

Sol. : Now,

$$f(z) = u + iv$$

$$i f(z) = i(u + iv) = iu - v$$

$$(1+i) f(z) = (u - v) + i(u + v)$$

$$F(z) = U + iV$$

So $U = u - v$ and $V = u + v$

$$U = u - v = e^x(\cos y - \sin y)$$

$$U_x = e^x(\cos y - \sin y)$$

$$U_y = e^x(-\sin y - \cos y)$$

$$U_{x(z,0)} = e^z(\cos 0 - \sin 0) = e^z$$

$$U_{y(z,0)} = -e^z$$

So $F(z) = \int U_{x(z,0)} dz - i \int U_{y(z,0)} dz$

$$= \int e^z dz + i \int e^z dz = e^z + ie^z$$

$$F(z) = (1+i) e^z$$

$$(1+i) f(z) = (1+i) e^z$$

$$f(z) = e^z + C$$

5.11.1 C - R Equation in Polar Co-ordinate

GTU : Winter-19, Maths-4, Marks 7

Now let $x = r \cos \theta$, $y = r \sin \theta$

$$r = \sqrt{x^2 + y^2} \quad \theta = \tan^{-1}\left(\frac{y}{x}\right)$$

Now we know that

$$f(z) = u + iv$$

$$z = r \cdot e^{i\theta} \text{ (Using polar co-ordinate)}$$

So $u + iv = f(r \cdot e^{i\theta}) \quad \dots(A)$

Differentiate equation (A) w.r. to r

$$\text{So } u_r + iv_r = f'(r \cdot e^{i\theta}) \cdot e^{i\theta} \quad \dots(1)$$

Differentiate equation (A) w.r. to θ

$$u_\theta + iv_\theta = f'(r \cdot e^{i\theta}) \cdot r \cdot e^{i\theta} \cdot i \quad \dots(2)$$

From equations (1) and (2) we get,

$$u_\theta + iv_\theta = \frac{(u_r + iv_r)}{e^{i\theta}} \cdot r \cdot e^{i\theta} \cdot i = ir u_r - r v_r$$

$$\text{So } u_\theta = -r v_r, \quad v_\theta = r v_r$$

$$\text{So } u_r = \frac{1}{r} v_\theta \text{ and } u_\theta = -r v_r$$

Ex. 5.11.19 Which of the following functions are analytic.

i) $f(z) = z^{1/2}$ ii) $f(z) = z^{3/2}$ iii) $f(z) = \sqrt[3]{r} e^{i\theta/2}$

iv) $f(z) = \frac{1}{z^4}$

v) $f(z) = e^{-\theta} \cos(\ln r) + i e^{-\theta} \sin(\ln r)$

GTU : Winter-23, CVNM, Marks 4
Winter-23, CVPPE, Marks 7

Sol. : i) $f(z) = z^{1/2}$

$$u + iv = (r \cdot e^{i\theta})^{1/2} = r^{1/2} \cdot e^{i\theta/2}$$

$$= r^{1/2} \left(\cos \frac{\theta}{2} + i \sin \frac{\theta}{2} \right)$$

$$u = r^{1/2} \cos \frac{\theta}{2} \quad v = r^{1/2} \sin \frac{\theta}{2}$$

$$u_r = \frac{1}{2\sqrt{r}} \cos \frac{\theta}{2} \quad v_r = \frac{1}{2\sqrt{r}} \sin \frac{\theta}{2}$$

$$u_\theta = -\frac{1}{2} r^{1/2} \sin \frac{\theta}{2} \quad v_\theta = \frac{1}{2} r^{1/2} \cos \frac{\theta}{2}$$

$$u_r = \frac{1}{r} v_\theta$$

$$u_\theta = -r v_r$$

$$\frac{1}{2\sqrt{r}} \cos \frac{\theta}{2} = \frac{1}{r} \cdot \frac{1}{2} r^{1/2} \cos \frac{\theta}{2} \quad \frac{-1}{2} r^{1/2} \sin \frac{\theta}{2} = -r \cdot \frac{1}{2\sqrt{r}} \sin \frac{\theta}{2}$$

C - R equation satisfied any where.

u_r, u_θ, v_r and v_θ continuous every where except $r = 0$ or $z = 0$.

So $f(z)$ is differentiable every where except $z = 0$.

So $f(z)$ is analytic every where except $z = 0$.

ii) $f(z) = z^{3/2}$

$$u + iv = (r \cdot e^{i\theta})^{3/2} = r^{3/2} e^{i3\theta/2}$$

$$= r^{3/2} \left(\cos \frac{3\theta}{2} + i \sin \frac{3\theta}{2} \right)$$

$$u = r^{3/2} \cos \frac{3\theta}{2}$$

$$v = r^{3/2} \sin \frac{3\theta}{2}$$

$$u_r = \frac{3}{2} r^{1/2} \cos \frac{3\theta}{2}$$

$$v_r = \frac{3}{2} r^{1/2} \sin \frac{3\theta}{2}$$

$$u_\theta = -\frac{3}{2} r^{3/2} \sin \frac{3\theta}{2}$$

$$v_\theta = \frac{3}{2} r^{3/2} \cos \frac{3\theta}{2}$$

Here $u_r = \frac{1}{r} v_\theta$ and $u_\theta = -r v_r$

So C-R equation satisfied any where.
 u_r, u_θ, v_r and v_θ continuous any where.

So $f(z)$ is differentiable any where.

So $f(z)$ is analytic any where.

iii) $f(z) = r^{1/3} e^{i\theta/2}$

$$u + iv = r^{1/3} \left(\cos \frac{\theta}{2} + i \sin \frac{\theta}{2} \right)$$

$$u = r^{1/3} \cos \frac{\theta}{2}$$

$$v = r^{1/3} \sin \frac{\theta}{2}$$

$$u_r = \frac{1}{3} r^{-2/3} \cos \frac{\theta}{2}$$

$$v_r = \frac{1}{3} r^{-2/3} \sin \frac{\theta}{2}$$

$$u_\theta = -\frac{1}{2} r^{1/3} \sin \frac{\theta}{2}$$

$$v_\theta = \frac{1}{2} r^{1/3} \cos \frac{\theta}{2}$$

$$u_r \neq \frac{1}{r} v_\theta \quad \text{and}$$

$$u_\theta \neq -rv_r$$

So C - R equation not satisfied.

So $f(z)$ is no where analytic.

$$\text{iv)} \quad f(z) = \frac{1}{z^4}$$

$$u + iv = z^{-4} = (r \cdot e^{i\theta})^{-4} = r^{-4} e^{-4i\theta}$$

$$= r^{-4} (\cos 4\theta - i \sin 4\theta)$$

$$u = r^{-4} \cos 4\theta$$

$$v = -r^{-4} \sin 4\theta$$

$$u_r = -4r^{-5} \cos 4\theta$$

$$v_r = 4r^{-5} \sin 4\theta$$

$$u_\theta = -4r^{-4} \sin 4\theta$$

$$v_\theta = -4r^{-4} \cos 4\theta$$

$$\text{Here} \quad u_r = \frac{1}{r} v_\theta \quad \text{and}$$

$$u_\theta = -rv_r$$

So C - R equation satisfied any where.

But u_r, u_θ, v_r and v_θ is not continuous at $r = 0$ (means $z = 0$)

So $f(z)$ is not differentiable only at $z = 0$.

So $f(z)$ is analytic any where except $z = 0$.

$$\text{v)} \quad f(z) = e^{-\theta} \cos(\ln r) + i e^{-\theta} \sin(\ln r)$$

$$u + iv = e^{-\theta} \cos(\ln r) + i e^{-\theta} \sin(\ln r)$$

$$u = e^{-\theta} \cos(\ln r)$$

$$v = e^{-\theta} \sin(\ln r)$$

$$u_r = -e^{-\theta} \sin(\ln r) \cdot \frac{1}{r}$$

$$u_\theta = -e^{-\theta} \cos(\ln r)$$

$$v_r = e^{-\theta} \cos(\ln r) \cdot \frac{1}{r}$$

$$v_\theta = -e^{-\theta} \sin(\ln r)$$

$$\text{So} \quad u_r = \frac{1}{r} v_\theta \quad \text{and} \quad u_\theta = -rv_r$$

Satisfied any where and u_r, u_θ, v_r and v_θ continuous any where.

So $f(z)$ is differentiable any where.

So $f(z)$ is analytic any where.

Ex. 5.11.20 State Cauchy-Riemann theorem. Write C-R equation in polar-form and verify it for $f(z) = \frac{z}{\bar{z}}$ in polar form.

GTU : Summer-18, CVNM, Marks 7

Sol. : C-R equation in polar form.

$$u_r = \frac{1}{r} v_\theta \quad \text{and} \quad u_\theta = -rv_r$$

$$\therefore f(z) = \frac{z}{\bar{z}}, \quad z = r \cdot e^{i\theta}, \quad \bar{z} = r \cdot e^{-i\theta}$$

$$u + iv = \frac{r \cdot e^{i\theta}}{r \cdot e^{-i\theta}} = e^{2i\theta}$$

$$u + iv = \cos 2\theta + i \sin 2\theta$$

$$u = \cos 2\theta \quad v = \sin 2\theta$$

$$u_r = 0 \quad v_r = 0$$

$$u_\theta = -2 \sin 2\theta \quad v_\theta = 2 \cos 2\theta$$

$$\text{Here, } u_r \neq \frac{1}{r} v_\theta \quad \text{and} \quad u_\theta \neq -rv_r$$

$\therefore$ C-R equation is not satisfied.

5.12 Harmonic Function

GTU : Winter-18, 19, 20, 21, 22, 23, 24,

Summer-18, 19, 20, 21, 22, 23, 24, Maths-4, Marks 3

A real valued function H at two variable x and y is said to be a Harmonic if H is satisfied Laplace equation.

$$H_{xx} + H_{yy} = 0$$

or

$$H_{xx}(x, y) + H_{yy}(x, y) = 0$$

Note : If a function $f(z) = u_{(x,y)} + iv_{(x,y)}$ is analytic in a domain D then its component functions u and v are harmonic in D .

Note : A function $f(z) = u + iv$ is analytic in a domain D iff v is a harmonic conjugate of u .

Ex. 5.12.1 Define harmonic function. Show that $u(x, y) = 2x - x^3 + 3xy^2$ is harmonic in some domain.

GTU : Summer-18, 19, Maths-4, Marks 3

Sol. : Harmonic function : A real valued function ϕ at two variable x and y is said to be a Harmonic if ϕ is satisfied Laplace equations.

$$\phi_{xx} + \phi_{yy} = 0$$

Now, $u(x, y) = 2x - x^3 + 3xy^2$

$$u_x = 2 - 3x^2 + 3y^2$$

$$u_{xx} = -6x \quad \dots(1)$$

and $u_y = 6xy$

$$u_{yy} = 6x \quad \dots(2)$$

From (1) and (2), we write,

$$u_{xx} + u_{yy} = -6x + 6x = 0$$

$\therefore u$ is Harmonic.

Ex. 5.12.2 Show that $f(x, y) = x^2 - y^2$ is Harmonic function.

**GTU : Winter-18,22, Marks 3
Summer-18, CVNM, Marks 2
Winter-24, CVNM, Marks 3
Summer-24, CVPDE, Marks 2**

Sol. : $f(x, y) = x^2 - y^2$

$$\therefore f_x = 2x \Rightarrow f_{xx} = 2,$$

and $f_y = -2y \Rightarrow f_{yy} = -2$

Now, $f_{xx} + f_{yy} = 2 - 2 = 0$

$\therefore f(x, y)$ is Harmonic function.

Ex. 5.12.3 Show that $u = y^3 - 3x^2y$ is Harmonic.

GTU : Summer-21, Winter-18, Maths-4, Marks 3

Sol. : $u = y^3 - 3x^2y$

$$u_x = -6xy \quad u_y = 3y^2 - 3x^2$$

$$u_{xx} = -6y \quad u_{yy} = 6y$$

$$\therefore u_{xx} + u_{yy} = -6y + 6y = 0$$

$\therefore u$ is Harmonic function.

Ex. 5.12.4 Show that $u = 2x - x^3 + 3xy^2$ is Harmonic in some domain.

**GTU : Summer-22, 24, Marks-4, Marks 3,
Summer-18,19, Winter-19,21, Marks 3,
Summer-20, 21, CVNM, Marks 3,
Winter-22, CVPDE, Marks 3,
Summer-23, VPDE, Marks 3**

Sol. : $u = 2x - x^3 + 3xy^2$

$$u_x = 2 - 3x^2 + 3y^2$$

$$\therefore u_{xx} = -6x$$

and $u_y = 6xy$

$$\therefore u_{yy} = 6x$$

Now, $u_{xx} + u_{yy} = -6x + 6x = 0$

$\therefore u$ is Harmonic function.

Ex. 5.12.5 Show that $u = x \sin x \cdot \cosh y - y \cos x \cdot \sinh y$ is harmonic.

GTU : Summer-22, CVNM, Marks 4

Sol. : $u = x \cdot \sin x \cdot \cosh y - y \cdot \cos x \cdot \sinh y$

Now, $u_x = \sin x \cdot \cosh y$

$$+ x \cos x \cdot \cosh y + y \sin x \cdot \sinh y$$

$$u_{xx} = \cos x \cdot \cosh y + \cos x \cdot \cosh y$$

$$- x \sin x \cdot \cosh y + y \cos x \cdot \sinh y$$

$\dots(1)$

and $u_y = x \sin x \cdot \sinh y - \cos x \cdot \sinh y$

$$- y \cos x \cdot \cosh y$$

$$u_{yy} = x \sin x \cosh y - \cos x \cdot \cosh y$$

$$- \cos x \cdot \cosh y - y \cos x \cdot \sinh y \dots (2)$$

From (1) and (2),

$$u_{xx} + u_{yy} = 0$$

$\therefore u$ is Harmonic.

Examples :

Ex. 5.12.6 Prove that $u_{(x,y)} = e^x \cos y$ is harmonic. Determine harmonic conjugate and the analytic function $f(z) = u + iv$.

GTU : Winter-23, CVNM, Marks 4
Winter-23, CVPDE, Marks 7

Sol. :

$$\begin{aligned}u_x &= e^x \cos y \\u_y &= -e^x \sin y \\u_{xx} &= e^x \cos y \\u_{yy} &= -e^x \cos y \\u_{xx} + u_{yy} &= 0\end{aligned}$$

So u is harmonic.

Now we find $v_{(x,y)}$.

Here, $f(z)$ is analytic function.

So C - R equation satisfied.

$$\begin{aligned}u_x &= v_y \text{ and } u_y = -v_x \\u_x &= e^x \cos y = v_y \\u_y &= -e^x \sin y = -v_x \\v_x &= e^x \sin y\end{aligned} \quad \dots(1)$$

Integrating equation (1) w.r. to x .

$$v_{(x,y)} = e^x \sin y + \phi(y) \quad \dots(2)$$

Now differentiate equation (2) w.r. to y .

$$v_y = +e^x \cos y + \phi'(y) \quad \dots(3)$$

$$\text{We know } v_y = e^x \cos y \quad \dots(4)$$

So from equations (3) and (4) (compare 3 and 4)

$$\text{We get } \phi'(y) = 0$$

$$\text{So } \phi(y) = c$$

$$\text{So } V_{(x,y)} = e^x \sin y + C$$

$$\text{Now } f(z) = u + iv = e^x \cos y + i(e^x \sin y + c).$$

Ex. 5.12.7 Find a harmonic conjugate of $u(x,y) = 2x - x^3 + 3xy^2$

GTU : Summer-18,19, Maths-4, Marks 4
Winter-18, Maths-4, Marks 4
Winter-19,21, Summer-22,24, Marks 4
Summer-20,21, CVNM, Marks 4
Summer-23, CVPDE, Marks 4
Winter-23, CVPDE, Marks 4

Sol. :

$$\begin{aligned}u(x,y) &= 2x - x^3 + 3xy^2 \\u_x &= 2 - 3x^2 + 3y^2 \\u_y &= 6xy\end{aligned}$$

Here, C-R equation is satisfied.

$$\therefore v_y = 2 - 3x^2 + 3y^2 \quad \dots(1)$$

$$v_x = -6xy \quad \dots(2)$$

Now integrating equation (1) w.r. to y .

$$\therefore v(x,y) = 2y - 3x^2y + y^3 + \phi(x) \quad \dots(3)$$

Now differentiate equation (3) w.r. to x .

$$v_x = -6xy + \phi'(x) \quad \dots(4)$$

Compare equation (2) and (4),

$$-6xy = -6xy + \phi'(x)$$

$$\therefore \phi'(x) = 0$$

$$\therefore \phi(x) = C$$

$$\therefore v(x,y) = 2y - 3x^2y + y^3 + C$$

Ex. 5.12.8 Show that the function

$u(x,y) = 3x^2y + 2x^2 - y^3 - 2y^2$ is Harmonic. Find the conjugate harmonic function v .

GTU : Winter-22, Maths-4, Marks 7
Winter-19, CVNM, Marks 7
Summer-19, CVNM, Marks 4

Sol. : Here, $u(x,y) = 3x^2y + 2x^2 - y^3 - 2y^2$

$$\therefore u_x = 6xy + 4x \quad \dots(1)$$

$$u_{xx} = 6y + 4$$

and $u_y = 3x^2 - 3y^2 - 4y \quad \dots(2)$

$$u_{yy} = -6y - 4$$

From (1) and (2),

$$u_{xx} + u_{yy} = 6y + 4 - 6y - 4 = 0$$

$\therefore u$ is Harmonic.

Now, C-R equation $u_x = v_y$ and $u_y = -v_x$

$$v_y = 6xy + 4x \quad \dots(3)$$

$$v_x = -3x^2 + 3y^2 + 4y \quad \dots(4)$$

Integrating equation (3) w.r. to y.

$$v(x, y) = \frac{6xy^2}{2} + 4xy + \phi(x)$$

$$v(x, y) = 3xy^2 + 4xy + \phi(x) \quad \dots(5)$$

Differentiate equation (5) w.r. to x.

$$v_x = 3y^2 + 4y + \phi'(x) \quad \dots(6)$$

Compare equation (4) and (6),

$$\therefore -3x^2 + 3y^2 + 4y = 3y^2 + 4y + \phi'(x)$$

$$\therefore -3x^2 = \phi'(x)$$

$$\therefore \phi(x) = \frac{-3x^3}{3} + C$$

$$\therefore v(x, y) = 3xy^2 + 4xy - x^3 + C$$

Ex. 5.12.9 Show that $u_{(x,y)}$ is harmonic in some domain D and find harmonic conjugate $v_{(x,y)}$ when

i) $u_{(x,y)} = y^3 - 3x^2y$ ii) $u_{(x,y)} = 2x(1-y)$

GTU : Winter-18,19, Summer-19,21, Maths-4, Marks 4

Sol. :

i) $u_{(x,y)} = y^3 - 3x^2y$

$$u_x = -6xy$$

$$u_y = 3y^2 - 3x^2$$

Here C - R equation is satisfied

$$v_y = -6xy \quad \dots(1)$$

$$v_x = 3x^2 - 3y^2 \quad \dots(2)$$

Now integrating equation (1) w.r. to y

$$v_{(x,y)} = \frac{-6xy^2}{2} + \phi(x) = -3xy^2 + \phi(x) \quad \dots(3)$$

Differentiate equation (3) w.r. to x

$$v_x = -3y^2 + \phi'(x) \quad \dots(4)$$

Compare equations (2) and (4).

$$3x^2 - 3y^2 = -3y^2 + \phi'(x)$$

We get $\phi'(x) = 3x^2$

$$\phi(x) = \frac{3x^3}{3} + C$$

So $v_{(x,y)} = -3xy^2 + x^3 + C$

$$f(z) = u + iv$$

$$= (y^3 - 3xy^2) + i(-3xy^2 + x^3 + C)$$

ii) $u_{(x,y)} = 2x(1-y)$

Now,

$$u_x = 2(1-y)$$

$$u_y = -2x$$

$$u_{xx} = 0$$

$$u_{yy} = 0$$

So $u_{xx} + u_{yy} = 0$

So u is harmonic.

C - R equation satisfied.

$$u_x = v_y \quad \text{and} \quad u_y = -v_x$$

$$v_y = 2(1-y) \quad \dots(1) \quad v_x = 2x \quad \dots(2)$$

Integrating equation (1) w.r.t. y

$$v_{(x,y)} = 2y - y^2 + \phi(x) \quad \dots(3)$$

Differentiate equations (3) w.r.to x

$$v_x = 0 + \phi'(x) \quad \dots(4)$$

Compare equations (2) and (4)

$$2x = \phi'(x)$$

So $\phi(x) = \frac{2x^2}{2} + C$

So.

$$V_{(x,y)} = 2y - y^2 + x^2 + C$$

Ex. 5.12.10 Given that $u(x, y) = x^2 - y^2$ and $v(x, y) = -\left(\frac{y}{x^2 + y^2}\right)$. Prove that u and v is Harmonic function but $u + iv$ is not an analytic function.

GTU : Winter-23, Maths-4, Marks 7

Sol. : First, we prove that u and v is Harmonic.

$$\therefore u = x^2 - y^2, \quad y_x = 2x \quad \text{and} \quad y_{xx} = 2$$

$$u_y = -2y \quad \text{and} \quad u_{yy} = 2$$

$$\therefore u_{xx} + u_{yy} = 2 - 2 = 0$$

$\therefore u$ is Harmonic.

and

$$v = \frac{-y}{x^2 + y^2}$$

$$v_x = \frac{y}{(x^2 + y^2)^2} \cdot 2x = \frac{2xy}{(x^2 + y^2)^2}$$

$$v_{xx} = \frac{2y(x^2 + y^2)^2 - 2xy \cdot 2(x^2 + y^2) \cdot 2x}{(x^2 + y^2)^4}$$

$$= \frac{(x^2 + y^2)[2y(x^2 + y^2) - 8x^2y]}{(x^2 + y^2)^4}$$

$$v_{xx} = \frac{2yx^2 + 2y^3 - 8x^2y}{(x^2 + y^2)^3}$$

$$= \frac{2y^3 - 6x^2y}{(x^2 + y^2)^3} \quad \dots(1)$$

Now,

$$v_y = \frac{(-1)(x^2 + y^2) - (-y)(2y)}{(x^2 + y^2)^2}$$

$$= \frac{-x^2 - y^2 + 2y^2}{(x^2 + y^2)^2}$$

$$v_y = \frac{y^2 - x^2}{(x^2 + y^2)^2}$$

$$\therefore v_{yy} = \frac{(2y)(x^2 + y^2)^2 - (y^2 - x^2)2(x^2 + y^2)(2y)}{(x^2 + y^2)^4}$$

$$= \frac{(x^2 + y^2)[2y(x^2 + y^2) - 4y(y^2 - x^2)]}{(x^2 + y^2)^4}$$

$$\therefore v_{yy} = \frac{2yx^2 + 2y^3 - 4y^3 + 4yx^2}{(x^2 + y^2)^3}$$

$$v_{yy} = \frac{6yx^2 - 2y^3}{(x^2 + y^2)^3} \quad \dots(2)$$

From (1) and (2),

$$v_{xx} + v_{yy} = \frac{2y^3 - 6yx^2}{(x^2 + y^2)^3} + \frac{6yx^2 - 2y^3}{(x^2 + y^2)^3} = 0$$

 $\therefore v$ is Harmonic.Now, we check $f(z) = u + iv$ is analytic.

$$u_x = 2x \quad v_x = \frac{2xy}{(x^2 + y^2)^2}$$

$$u_y = -2y \quad v_y = \frac{y^2 - x^2}{(x^2 + y^2)^2}$$

Here, C-R equation $u_x = v_y$ and $u_y = -v_x$ is not satisfied. $\therefore f(z) = u + iv$ is not analytic.**Milne Thomson's method :**

$$\text{Let } f(z) = u_{(x,y)} + iv_{(x,y)}$$

$$f'(z) = u_x + iv_x$$

Using C - R equation $u_x = v_y$ and $u_y = -v_x$

$$f'(z) = u_x - iv_y$$

$$f'(z) = u_{x(z,0)} - iv_{y(z,0)}$$

Taking both side integration,

$$\int f'(z)dz = \int u_{x(z,0)}dz - i \int u_{y(z,0)}dz$$

Note : If given that real part of $f(z)$ and we want to find analytic function $f(z)$ then we use,

$$f(z) = \int u_{x(z,0)}dz - i \int u_{y(z,0)}dz + u(0,0)$$

Note : If given that imaginary part of $f(z)$ and we want to find analytic function $f(z)$ then we use,

$$f(z) = \int v_{y(z,0)}dz + i \int v_{x(z,0)}dz + v(0,0)$$

Ex. 5.12.11 Find the analytic function of which real part is $e^{-x}((x^2 - y^2) \cos y + 2xy \sin y)$ **Sol. :** $u = e^{-x}((x^2 - y^2) \cos y + 2xy \sin y)$

$$u_{(0,0)} = 0$$

$$u_x = -e^{-x}((x^2 - y^2) \cos y + 2xy \sin y)$$

$$+ e^{-x}(2x \cos y + 2y \sin y)$$

$$u_y = e^{-x}(-2y \cos y + (x^2 - y^2)(-\sin y))$$

$$+ 2x \sin y + 2xy \cos y$$

$$u_{x(z,0)} = -e^{-z}(z^2) + e^{-z}(2z)$$

$$u_{y(z,0)} = e^{-z}(0) = 0$$

$$\begin{aligned}
 f(z) &= \int u_{x(z,0)} dz - i \int v_{y(z,0)} dz \\
 &= \int -e^{-z}(z^2) + e^{-z}(2z) dz - i \int 0 dz \\
 &= \int e^{-z}(2z - z^2) dz \\
 &= [(2z - z^2)(-e^{-z}) - (2 - 2z)(e^{-z}) \\
 &\quad + (-2)(-e^{-z})] + C \\
 &= [e^{-z}(-\frac{1}{2}z + z^2 - \frac{1}{2} + \frac{1}{2}z + \frac{1}{2})] + C \\
 f(z) &= e^{-z} \cdot z^2 + C
 \end{aligned}$$

Ex. 5.12.12 Construct the analytic function $f(z)$ when

i) $u = x^3 - 3xy^2 + 3x + 1$

ii) $u = y^3 - 3x^2y$ iii) $u = \sin x \cosh y$

Sol. : i)

$$u = x^3 - 3xy^2 + 3x + 1 \Rightarrow u(0,0) = 1$$

$$u_x = 3x^2 - 3y^2 + 3 \Rightarrow u_{x(z,0)} = 3z^2 + 3$$

$$u_y = -6xy \Rightarrow u_{y(z,0)} = 0$$

$$f(z) = \int u_{x(z,0)} dz - i \int u_{y(z,0)} dz$$

$$= \int 3z^2 + 3 dz - i \int 0 dz$$

$$= \frac{3z^3}{3} + 3z + C + u(0,0)$$

$$f(z) = z^3 + 3z + C + 1$$

ii)

$$u = y^3 - 3x^2y$$

$$u_x = -6xy \Rightarrow u_{x(z,0)} = 0$$

$$u_y = 3y^2 - 3x^2 \Rightarrow u_{y(z,0)} = -3z^2$$

$$f(z) = \int u_{x(z,0)} dz - i \int u_{y(z,0)} dz$$

$$= \int 0 dz - i \int -3z^2 dz$$

$$= +i(z^3) + C + y(0,0)$$

$$= iz^3 + C$$

iii)

$$u = \sin x \cosh y$$

$$u_x = \cos x \cosh y \Rightarrow u_{x(z,0)} = \cos z$$

$$u_z = \sin x \sinh y \Rightarrow u_{y(z,0)} = 0$$

$$u(0,0) = 0$$

$$f(z) = \int u_{x(z,0)} dz - i \int u_{y(z,0)} dz$$

$$= \int \cos z dz - i \int 0 dz$$

$$= \sin z + C + u(0,0)$$

$$f(z) = \sin z + C$$

Ex. 5.12.13 Determine the analytic function $f(z) = u + iv$ whose real part is $u = x^2 - y^2$.

GTU : Summer-18, Maths-4, Marks 4
Summer-23, Maths-4, Marks 7

Sol. : Here, $u = x^2 - y^2$ is given

$$u = x^2 - y^2 \Rightarrow u(0,0) = 0$$

$$u_x = 2x \Rightarrow u_{x(z,0)} = 2z$$

$$u_y = -2y \Rightarrow u_{y(z,0)} = 0$$

Using Milne Thomson's Method,

$$f(z) = \int u_{x(z,0)} dz - i \int u_{y(z,0)} dz + u(0,0)$$

$$= \int 2z dz - i \int 0 dz + 0$$

$$= \frac{2z^2}{2} + C$$

$$f(z) = z^2 + C$$

Ex. 5.12.14 Show that the function

$u(x, y) = x^4 - 6x^2y^2 + y^4$ is Harmonic. Also find the analytic function $f(z) = u(x, y) + i v(x, y)$.

GTU : Winter-23, Maths-4, Marks 7

Sol. : Here, $u = x^4 - 6x^2y^2 + y^4$

$$\therefore u_x = 4x^3 - 12xy^2$$

$$u_{xx} = 12x^2 - 12y^2 \quad \dots(1)$$

and $u_y = -12x^2y + 4y^3$

$$u_{yy} = -12x^2 + 12y^2 \quad \dots(2)$$

From (1) and (2),

$$u_{xx} + u_{yy} = 12x^2 - 12y^2 - 12x^2 + 12y^2$$

$\therefore u$ is Harmonic.

Now, we find analytic function $f(z)$ using Milne Thomson's Method.

$$f(z) = \int u_{x(z,0)} dz - i \int u_{y(z,0)} dz + u(0,0)$$

$$\therefore u(0,0) = 0$$

$$u_x = 4x^3 - 12xy^2 \Rightarrow u_{x(z,0)} = 4z^3$$

$$u_y = -12x^2y + 4y^3 \Rightarrow u_y(z, 0) = 0$$

$$\begin{aligned}\therefore f(z) &= \int 4z^3 dz - i \int 0 dz + 0 \\ &= \frac{4z^4}{4} + C\end{aligned}$$

$$\therefore f(z) = z^4 + C$$

Ex. 5.12.15 Determine the analytic function whose real part is $e^{-x}(x \cos y - y \sin y)$, using Milne - Thomson's Method.

GTU : Winter-24, Maths-4, Marks 7

Sol. : $u(x, y) = e^{-x}(x \cos y - y \sin y)$

Milne-Thomson's Method,

$$f(z) = \int u_x(z, 0) dz - i \int u_y(z, 0) dz + u(0, 0)$$

$$\therefore u(0, 0) = 0$$

$$u_x = -e^{-x}(x \cos y - y \sin y) + e^{-x} \cos y$$

$$\Rightarrow u_x(z, 0) = -e^{-z}(z) + e^{-z} 1$$

$$u_y = e^{-x}(x(-\sin y) - (\sin y + y \cos y))$$

$$\Rightarrow u_y(z, 0) = e^{-z}(z(0) - (0 + 0)) = 0$$

$$\begin{aligned}\therefore f(z) &= \int -e^{-z}z + e^{-z} dz - i \int 0 dz + 0 \\ &= \int e^{-z}(1 - z) dz \\ &= (1 - z) \int e^{-z} dz - \int ((-1) \int e^{-z} dz) dz \\ &= (1 - z)(-e^{-z}) + \int (-e^{-z}) dz \\ &= -e^{-z} + ze^{-z} + e^{-z} + C\end{aligned}$$

$$f(z) = z \cdot e^{-z} + C$$

Ex. 5.12.16 If $u = x^2 - y^2$, find the corresponding analytic function $f(z) = u + iv$.

GTU : Winter-20, Maths-4, Marks 4

Summer-18, CVNM, Marks 2

Winter-23, CVNM, Marks 3

Summer-24, CVPDE, Marks 2

Sol. : $\Rightarrow u = x^2 - y^2$

$$\therefore u_x = 2x \Rightarrow u_x(z, 0) = 2z$$

$$u_y = -2y \Rightarrow u_y(z, 0) = 0$$

$$u(0, 0) = 0$$

Now, using Milne - Thomson's Method,

$$\begin{aligned}f(z) &= \int u_x(z, 0) dz - i \int u_y(z, 0) dz + u(0, 0) \\ &= \int 2z dz - i \int 0 dz + 0\end{aligned}$$

$$f(z) = \frac{2z^2}{2} + C$$

$$\therefore f(z) = z^2 + C$$

Ex. 5.12.17 Determine the analytic function whose imaginary part is $e^x(x \cos y - y \sin y)$.

GTU : Winter-22, Maths-4, Marks 7

Sol. : Here, Important part $v = e^x(x \cos y - y \sin y)$ is given,

Here, we use Milne - Thomson's method,

$$f(z) = \int v_y(z, 0) dz + i \int v_x(z, 0) dz + v(0, 0)$$

$$v = e^x(x \cos y - y \sin y)$$

$$v(0, 0) = 0$$

$$v_x = e^x(x \cos y - y \sin y) + e^x(\cos y)$$

$$\Rightarrow v_x(z, 0) = e^z(z) + e^z$$

$$v_y = e^x(-x \sin y - (\sin y + y \cos y))$$

$$\Rightarrow v_y(z, 0) = 0$$

$$f(z) = 0 + i \int e^z z + e^z dz + 0$$

$$f(z) = i[z e^z - e^z + e^z + C]$$

$$f(z) = i[z e^z + C]$$

Ex. 5.12.18 Find an analytic function $f(z) = u + iv$ whose real part is $u = x^4 - 6x^2y^2 + y^4$. Also find the corresponding function v .

GTU : Summer-22, CVNM, Marks 7

Winter-22, CVNM, Marks 3

Sol. : $\Rightarrow u = x^4 - 6x^2y^2 + y^4$

Here, we use Milne-Thomson's Method,

$$f(z) = \int u_x(z, 0) dz - i \int u_y(z, 0) dz + u(0, 0)$$

$$u(0, 0) = 0$$

$$u_x = 4x^3 - 12xy^2 \Rightarrow u_x(z, 0) = 4z^3$$

$$u_y = -12x^2y + 4y^3 \Rightarrow u_y(z, 0) = 0$$

$$f(z) = \int 4z^3 dz - i \int 0 dz + 0$$

$$f(z) = z^4 + C$$

$$u + iv = (x + iy)^4$$

$$= x^4 + 4x^3(iy) + 6x^2(iy)^2 + 4x(iy)^3 + y^4$$

$$= x^4 + i4x^3y - 6x^2y^2 - i4xy^3 + y^4$$

$$= (x^4 - 6x^2y^2 + y^4) + i(4x^3y - 4xy^3)$$

$$\therefore v = 4x^3y - 4xy^3 \text{ and } u = x^4 - 6x^2y^2 + y^4$$

Ex. 5.12.19 Show that $u(x, y) = x^2 - y^2 + x$ is Harmonic.

Find the corresponding analytic function $f(z) = u + iv$.

GTU : Winter-24, CVPDE, Marks 4

$$\text{Sol. : } \Rightarrow u(x, y) = x^2 - y^2 + x$$

$$u_x = 2x + 1, u_{xx} = 2$$

$$u_y = -2y, u_{yy} = -2$$

$$\therefore u_{xx} + u_{yy} = 2 - 2 = 0$$

$\therefore u$ is Harmonic.

$$\text{Now, } u_x(z, 0) = 2z + 1 \quad \left| \quad u(0, 0) = 0 \right.$$

$$u_y(z, 0) = 0$$

By Milne-Thomson's Method,

$$f(z) = \int u_x(z, 0) dz - i \int u_y(z, 0) dz + u(0, 0)$$

$$= \int 2z + 1 dz - i \int 0 dz + 0$$

$$= \frac{2z^2}{2} + z + C$$

$$f(z) = z^2 + z + C$$

Ex. 5.12.20 Find the constant a so that

$u(x, y) = ax^2 - y^2 + xy$ is harmonic. Find its harmonic conjugate and hence $f(z) = u + iv$

$$\text{Sol. : } u = ax^2 - y^2 + xy$$

$$u_x = 2ax + y \quad u_{xx} = 2a$$

$$u_y = -2y + x \quad u_{yy} = -2$$

$$\text{So } u_{xx} + u_{yy} = 0$$

$$2a - 2 = 0$$

$$(a - 1) = 0 \Rightarrow a = 1$$

$$u = x^2 - y^2 + xy$$

$$u_x = 2x - y \quad u_y = -2y + x$$

$$u_{x(z,0)} = 2z \quad u_{y(z,0)} = z$$

$$v = \int 2z dz - i \int z dz = z^2 - iz^2 + C$$

$$= \left(1 - \frac{i}{2}\right) z^2 + C = \left(1 - \frac{i}{2}\right) (x + iy)^2 + C$$

$$= \left(1 - \frac{i}{2}\right) (x^2 + 2ixy - y^2) + C$$

$$= x^2 + 2ixy - y^2 - \frac{i}{2} x^2 - xy + \frac{i}{2} y^2$$

$$= (x^2 - y^2 - xy) + i \left(2xy - \frac{x^2}{2} + \frac{y^2}{2} \right)$$

Ex. 5.12.21 If $W = \phi + i\psi$ represent the complex potential for an electric field and

$$\psi = x^2 - y^2 + \frac{x}{x^2 y^2} \text{ determine the function } \phi.$$

$$\text{Sol. : } \psi = x^2 - y^2 + \frac{x}{x^2 - y^2}$$

$$\psi_x = 2x + \frac{(x^2 + y^2) - x(2x)}{(x^2 + y^2)^2}$$

$$\psi_{x(z,0)} = 2z + \frac{(-z^2)}{z^4} = 2z - \frac{1}{z^2}$$

$$\psi_y = -2y + \frac{(0) - x(2y)}{(x^2 + y^2)^2}$$

$$\psi_{y(z,0)} = 0$$

$$f(z) = \int \psi_y(z, 0) dz + i \int \psi_x(z, 0) dz$$

$$= 0 + i \int 2z - \frac{1}{z^2} dz = z^2 + \frac{1}{2} + C$$

$$= (x + iy)^2 + \frac{(x - iy)}{x^2 + y^2}$$

$$= \frac{(x^2 + 2ixy - y^2)(x^2 + y^2) + x - iy}{x^2 - y^2}$$

$$\begin{aligned}
 &= \frac{x^4 + x^2 y^2 + 2ix^3 y + 2ixy^3}{x^2 + y^2} \\
 &= \frac{-x^2 - y^4 + x - iy}{x^2 + y^2} \\
 &= \frac{(x^4 - y^4) + ix}{x^2 + y^2} + i \left(\frac{2x^3 y + 2xy^3}{x^2 + y^2} \right) \\
 &= x^2 - y^2 + \frac{x}{x^2 + y^2} + i \left(2xy - \frac{y}{x^2 + y^2} \right) \\
 \phi &= 2xy - \frac{y}{x^2 + y^2} + C
 \end{aligned}$$

Ex. 5.12.22 Find the analytic function $f(z)$ whose real part is $u = ax^3 + bxy$ and also find a and b where u is harmonic.

GTU : Summer-23, Maths-4, Marks 7
Winter-23, CVPDE, Marks 3

Sol. : $u = ax^3 + bxy \Rightarrow u_x = 3ax^2 + by$

$u_y = bx \quad u_{xx} = 6ax$

$u_{yy} = 0$

$u_{xx} + u_{yy} = 6ax + 0 = 0$

$\therefore 6ax = 0$

$a = 0 \text{ and } b \in \mathbb{C}$

So $u = bxy, \quad u_x = by, \quad u_y = bx$

$u_{x(z,0)} = u_{y(z,0)} = bz$

$f(z) = \int u_{x(z,0)} dz - i \int u_{y(z,0)} dz$

$= \int 0 dz - i \int bz dz$

$f(z) = -i \frac{bz^2}{2} + C$

Ex. 5.12.23 Show that on analytic function with constant modulus is constant.

GTU : Summer-21, Maths-4, Marks 7

Sol. : Let

$f(z) = u + iv$

$|f(z)| = C$

$u^2 + v^2 = C \quad \dots(1)$

Differentiate equation (1) w.r.to x

$2u \frac{\partial u}{\partial x} + 2v \frac{\partial v}{\partial x} = 0$

$2u \frac{\partial u}{\partial y} + 2v \frac{\partial v}{\partial y} = 0$

Using C-R equation

$2u u_x - 2v u_y = 0 \quad \dots (2)$

$(\because u_x = u_y \text{ and } u_y = -v_x)$

Multiply equation (1) w.r. to v and

Multiply equation (2) w.r. to u

$2v u_x + 2u u_y = 0 \quad \dots (3)$

$2uv u_x - 2v^2 u_y = 0 \quad \dots (4)$

$2uv u_x + 2u^2 u_y = 0 \quad \dots (5)$

$-(2u^2 + 2v^2) u_y = 0$

$(u^2 + v^2) \frac{\partial u}{\partial y} = 0$

$\frac{\partial u}{\partial y} = 0 \Rightarrow u = C$

Similarly $\frac{\partial v}{\partial x} = 0 \Rightarrow v = C$

So $f(z) = u + iv$ is also constant.

Ex. 5.12.24 Let a function $f(z)$ be analytic in a domain D . Prove that $f(z)$ must be constant in D in each of the following cases :

a) If $f(z)$ is real valued for all z in D .

b) If $\overline{f(z)}$ is analytic in D .

Sol. :

a) Here given that $f(z)$ is real valued function for all z in D .

So, $f(z) = u + iv$ but $v = 0$

So, $v_x = 0 \quad v_y = 0$

$f(z) = u + iv$ is analytic

C-R equation is satisfied.

$u_x = v_y \text{ and } u_y = -v_x$

$$u_x = 0 \text{ and } u_y = 0$$

So,

$$u_x = u_y = 0 \Rightarrow u = C$$

$$v_x = v_y = 0 \Rightarrow v = C$$

b) Here $f(z) = u + iv$ and $\overline{f(z)} = u - iv$ is analytic.

Now we take first

$$f(z) = u + iv \text{ is analytic.}$$

So C-R equation satisfied.

$$u_x = v_y \quad \dots(1)$$

$$\text{and } u_y = -v_x \quad \dots(2)$$

$$\overline{f(z)} = u - iv \text{ is analytic.}$$

So C-R equation satisfied

$$u_x = -v_y \quad \dots(3)$$

$$u_y = v_x \quad \dots(4)$$

From equations (1) and (3) we get,

$$u_x + u_x = 0 \Rightarrow 2u_x = 0$$

$$\Rightarrow u_x = 0$$

$$\Rightarrow u = C$$

From equations (2) and (4) we get

$$u_y - u_y = -v_x - v_x$$

$$-2v_x = 0 \Rightarrow v_x = 0 \Rightarrow v = C$$

So, $f(z) = u + iv$ is also constant.

Ex. 5.12.25 Show that an analytic function in a domain with its derivative zero for every point of the domain is constant.

Sol. : We have to show that if $f'(z)$ is zero for every point of a domain D , then $f(z)$ is constant in D .

We have for such function,

$$f(z) = u + iv \quad f'(z) = u_x + iv_x$$

So that $f'(z) = 0$

$$\Rightarrow u_x = 0 \text{ and } v_x = 0$$

$$\Rightarrow v_y = 0 \text{ and } u_y = 0$$

$$\Rightarrow u_x = 0, u_y = 0 \Rightarrow u \text{ is constant.}$$

$$\text{ans } v_x = 0, v_y = 0 \Rightarrow v \text{ is constant.}$$

Ex. 5.12.26 If $f(z)$ is regular function (Analytic function) of z then

Prove that :

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) | \text{Rel}(f(z)) |^2 = 2 | f'(z) |^2$$

GTU : Summer-22, Maths-4, Marks 7

Sol. : We have

$$f(z) = u + iv$$

$$\text{Rel}(f(z)) = u$$

$$| \text{Rel}(f(z)) |^2 = | u^2 | = u^2$$

$$\text{L.H.S} = \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) (u^2)$$

$$= \frac{\partial^2}{\partial x^2} (u^2) + \frac{\partial^2}{\partial y^2} (u^2)$$

$$= \frac{\partial}{\partial x} (2u u_x) + \frac{\partial}{\partial y} (2u u_y)$$

$$= 2u_x^2 + 2u u_{xx} + 2u_y^2 + 2u u_{yy}$$

$$= 2u(u_{xx} + u_{yy}) + 2(u_x^2 + u_y^2)$$

But we know that $f(z)$ is analytic then $f(z)$ is also harmonic so.

$$u_{xx} + u_{yy} = 0$$

$$= 2u(0) + 2(u_x^2 + u_y^2) = 2(u_x^2 + u_y^2)$$

$$= 2 | f'(z) |^2$$

$$(\because f(z) = u + iv)$$

$$f'(z) = u_x + iv_x$$

$$| f'(z) | = \sqrt{u_x^2 + v_x^2} = \sqrt{u_x^2 + u_y^2}$$

Ex. 5.12.27 If $f(z) = u + iv$ is regular function (analytic function) then prove that

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) | I_m(z) |^2 = 2 | f'(z) |^2$$

GTU : Winter-23, CVNM, Marks 4
Winter-23, CVPDE, Marks 7

$$\text{Sol. : } \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) (v^2)$$

$$\begin{aligned}
 &= \frac{\partial^2}{\partial x^2}(v^2) + \frac{\partial^2}{\partial y^2}(v^2) \\
 &= \frac{\partial}{\partial x}(2v v_x) + \frac{\partial}{\partial y}(2v v_y) \\
 &= 2v_x^2 + 2v v_{xx} + 2v_y^2 + 2v v_{yy} \\
 &= 2v(v_{xx} + v_{yy}) + 2v_x^2 + 2v_y^2 \\
 &= 2v(0) + 2(v_x^2 + v_y^2) = 2(v_x^2 + v_y^2) \\
 &= 2|f'(z)|^2
 \end{aligned}$$

Ex. 5.12.28 If $f(z) = u + iv$ is analytic function then prove that

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right) |f(z)|^2 = 4|f'(z)|^2$$

GTU : Summer-19, Maths-4, Marks 7

Sol. : $|f(z)|^2 = f(z) \cdot \overline{f(z)}$

$$\begin{aligned}
 &= \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right) |f(z)|^2 \\
 &= \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right) (u^2 + v^2) \\
 &= \frac{\partial^2}{\partial x^2} (u^2 + v^2) + \frac{\partial^2}{\partial y^2} (u^2 + v^2) \\
 &= \frac{\partial}{\partial x} (2u u_x + 2v v_x) + \frac{\partial}{\partial y} (2u u_y + 2v v_y) \\
 &= 2u_x^2 + 2u u_{xx} + 2v_x^2 + 2v v_{xx} \\
 &\quad + 2u_y^2 + 2u u_{yy} + 2v_y^2 + 2v v_{yy} \\
 &= 2(u_x^2 + v_x^2) + 2(u_y^2 + v_y^2) \\
 &\quad + 2u(u_{xx} + u_{yy}) + 2v(v_{xx} + v_{yy}) \\
 &= 2(u_x^2 + v_x^2) + 2(u_y^2 + v_y^2) \\
 &\quad (\because u_{xx} + u_{yy} = 0) \\
 &\quad v_{xx} + v_{yy} = 0 \\
 &= 2|f'(z)|^2 + 2|f'(z)|^2 = 4|f'(z)|^2
 \end{aligned}$$

Ex. 5.12.29 Prove that $\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right) = 4 \frac{\partial^2}{\partial z \partial \bar{z}}$

Sol. : Let $x + iy = z$, $x - iy = \bar{z}$

$$x = \frac{1}{2}(z + \bar{z}), \quad y = -\frac{i}{2}(z - \bar{z})$$

$$\frac{\partial x}{\partial z} = \frac{1}{2} \frac{\partial x}{\partial z} = \frac{1}{2} \frac{\partial y}{\partial z} = \frac{-i}{2}, \quad \frac{\partial y}{\partial z} = \frac{i}{2}$$

$$\frac{\partial}{\partial z} = \frac{\partial}{\partial x} \cdot \frac{\partial x}{\partial z} + \frac{\partial}{\partial y} \cdot \frac{\partial y}{\partial z} = \frac{\partial}{\partial x} \left(\frac{1}{2}\right) + \frac{\partial}{\partial y} \left(-\frac{i}{2}\right)$$

$$\frac{\partial}{\partial \bar{z}} = \frac{\partial}{\partial x} \cdot \frac{\partial x}{\partial \bar{z}} + \frac{\partial}{\partial y} \cdot \frac{\partial y}{\partial \bar{z}} = \frac{\partial}{\partial x} \left(\frac{1}{2}\right) + \frac{\partial}{\partial y} \left(\frac{i}{2}\right)$$

Now, $\frac{\partial^2}{\partial z \partial \bar{z}} = \frac{\partial}{\partial z} \cdot \frac{\partial}{\partial \bar{z}} = \frac{1}{2} \left(\frac{\partial}{\partial x} - i \frac{\partial}{\partial y} \right) \frac{1}{2} \left(\frac{\partial}{\partial x} + i \frac{\partial}{\partial y} \right)$

$$= \frac{1}{4} \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right)$$

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) = 4 \frac{\partial^2}{\partial z \partial \bar{z}}$$

Ex. 5.12.30 If $w = f(z)$ is a regular function at z prove

that $\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right) \log |f'(z)| = 0$

Sol. : $\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right) \{\log |f'(z)|\}$

$$= 4 \frac{\partial^2}{\partial z \partial \bar{z}} \{\log |f'(z)|\}$$

$$= 4 \frac{\partial^2}{\partial z \partial \bar{z}} \left\{ \frac{1}{2} \log |f'(z)|^2 \right\}$$

$$= 2 \frac{\partial^2}{\partial z \partial \bar{z}} \{ (f'(z) \overline{f'(z)}) \}$$

$$= 2 \frac{\partial^2}{\partial z \partial \bar{z}} \{ \log f'(z) + \log \overline{f'(z)} \}$$

$$= 2 \frac{\partial}{\partial \bar{z}} \left\{ \frac{f''(z)}{f'(z)} \right\} = 0$$

□□□